

Assessing the Performance of Clustering Methods in Spatially Resolved Transcriptomics Data

Xiaoying Liao

A thesis submitted in partial fulfillment of
the requirements for the degree of
Bachelor of Science (Honours)

School of Mathematics and Statistics



October 2023

Statement of originality

This is to certify that to the best of my knowledge, the content of this thesis is my own work. This thesis has not been submitted for any degree or other purposes.

I certify that the intellectual content of this thesis is the product of my own work and that all the assistance received in preparing this thesis and sources have been acknowledged.

Xiaoying Liao

Abstract

Spatially resolved transcriptomics technologies enable the measurement of transcriptome information while preserving the spatial context. Although spatial clustering remains relatively unknown in this field, there is an emerging need for a comprehensive assessment of clustering methods. This study provides an in-depth analysis of various clustering methods, ranging from classical single-cell RNA-sequencing (scRNA-seq) clustering techniques to contemporary spatial clustering approaches incorporating 2D and 3D coordinates. Employing simulated and real data, we evaluate the performance of various spatial clustering methods. Our results suggest that certain spatial algorithms incorporating spatial coordinates outperform classical clustering methods in certain scenarios. Moreover, 3D spatial clustering methods demonstrate significant advantages over 2D algorithms regarding spatial domain detection accuracy. However, the inconsistent performance of certain algorithms highlights the complexity of drawing definitive conclusions. This study not only reveals the inherent capabilities and limitations of spatial clustering techniques but also lays the groundwork for future exploration of spatial domain detection in spatially resolved transcriptomics data.

Keywords: scRNA-seq ; spatially resolved transcriptomics ; spatial clustering ; spatial domain detection ; benchmarking

Acknowledgements

This Honours year has been a challenging year for me. Not only have I grown rapidly in my academic abilities but also in my resilience to stress. My thesis journey ends in chapter six, but life's journey continues. Looking back, I have faced challenges and spent many sleepless nights understanding complicated ideas. However, with the support of supervisors, mentors, friends, and family, I overcame and grew stronger. These experiences taught me that knowledge is more than just facts. It lights our way during tough times. With a grateful heart, I look to new horizons and embrace the future, strengthened by the invaluable lessons of this monumental chapter in my life.

Firstly, I would like to express my deepest gratitude to my supervisor, Prof. Jean Yang, for her support, guidance, and profound insights throughout this year. In my thesis, she provided invaluable suggestions and patiently pointed out areas that needed improvement and enhancement. I am deeply grateful to her for teaching me how to effectively structure my thesis and how to create a cohesive story when writing the thesis. These lessons are invaluable treasures in my future academic career.

I am also extremely grateful to my co-supervisor, Dr. Clara Grazian. She provided invaluable advice and feedback on my thesis and always supported and encouraged me, especially during difficult times. She patiently guided me in clarifying my thoughts and structure for my Honours project and presentation, boosting my confidence in the process.

I would like to express my sincere appreciation to my mentors, Dr. Yue Cao and Dr. Yingxin Lin, for their patient guidance. They always helped me when I encountered coding and modelling problems and provided me with the greatest support.

I would also like to thank Dr. Shila Ghazanfar and Dr. Ellis Patrick for their invaluable advice on spatial transcriptomics and valuable insights and recommendations on the thesis content.

I would like to thank everyone in the Sydney Precision Bioinformatics Alliance for providing valuable suggestions for my project and Honours presentation. In particular, I am very grateful to Lijia Yu and Carissa Chen for their guidance on the ideas and content of my thesis. I would also like to thank Cabiria Liang for helping me run the simulation model. I am very grateful to be able to study with my friends in the School of Mathematics and Statistics. In particular, I would like to thank Dai Zuocheng, Guan Gui, and Peikang Lu for their valuable advice on my Honours presentation.

I would also like to extend my gratitude to Dr. Xi Xiao, who not only provided invaluable advice and insights but also generously took the time to proofread my thesis.

Finally, I would like to thank my parents for their continuous support and encouragement, which enabled me to complete this Honours year successfully.

Contents

1	Introduction	1
1.1	Biological background	1
1.2	Single-cell technologies and data structures	2
1.2.1	Data structures comparison	2
1.2.2	Single-cell RNA sequencing technology	3
1.2.3	Two-dimensional spatially resolved transcriptomics	4
1.2.4	Three-dimensional spatially resolved transcriptomics	7
1.3	Data Science challenges	8
1.3.1	Common challenges in single-cell RNA-seq data	8
1.3.2	Unique challenges in spatial transcriptomics	9
1.4	The aim of the thesis	9
1.5	Outline of thesis	10
2	Literature review on clustering analysis in single-cell field	11
2.1	Single-cell RNA sequencing clustering analysis	11
2.2	Clustering methods for scRNA-seq data	14
2.2.1	K-means based clustering methods	14
2.2.2	Hierarchical based clustering methods	15
2.2.3	Community detection based clustering methods	15
2.2.4	Summary clustering methods for scRNA-seq data	16
2.2.5	Challenges in scRNA-seq clustering analysis	16
2.3	Spatial clustering methods for spatially resolved transcriptomics data	18
2.3.1	2D spatial clustering methods	21
2.3.2	3D spatial clustering methods	31
2.3.3	Challenges in spatial clustering analysis	34
2.4	Summary	34
3	Two-dimensional Spatial Domain Detection	35
3.1	Data description	35
3.2	Experimental design	35
3.3	Simulation studies	36
3.3.1	Simulation methods: CARD and scDesign3	36
3.3.2	Input patterns	39
3.3.3	Evaluation procedure	41
3.3.4	Results	44
3.4	Summary	48

4	2D vs 3D Spatial Clustering Comparison	49
4.1	Data description	49
4.2	3D reconstruction	50
4.3	Experimental design	51
4.4	Results	52
4.5	Summary	54
5	Conclusion	55
5.1	Discussion	55
5.2	Conclusion	56
6	Appendix	57
6.1	Data availability	57
6.2	Code availability	57
6.3	Supplementary information	58

Chapter 1

Introduction

1.1 Biological background

The cell is regarded as the fundamental unit of biology, and all multicellular organisms have a variety of unique cell types (Kiselev et al., 2019). There are many types of cells in multicellular organisms, including humans. As shown in Figure 1.1, there are different cell types in various tissue structures in the human body (Karlsson et al., 2021). For example, there are red and white blood cells in the blood (Bain, 2017).

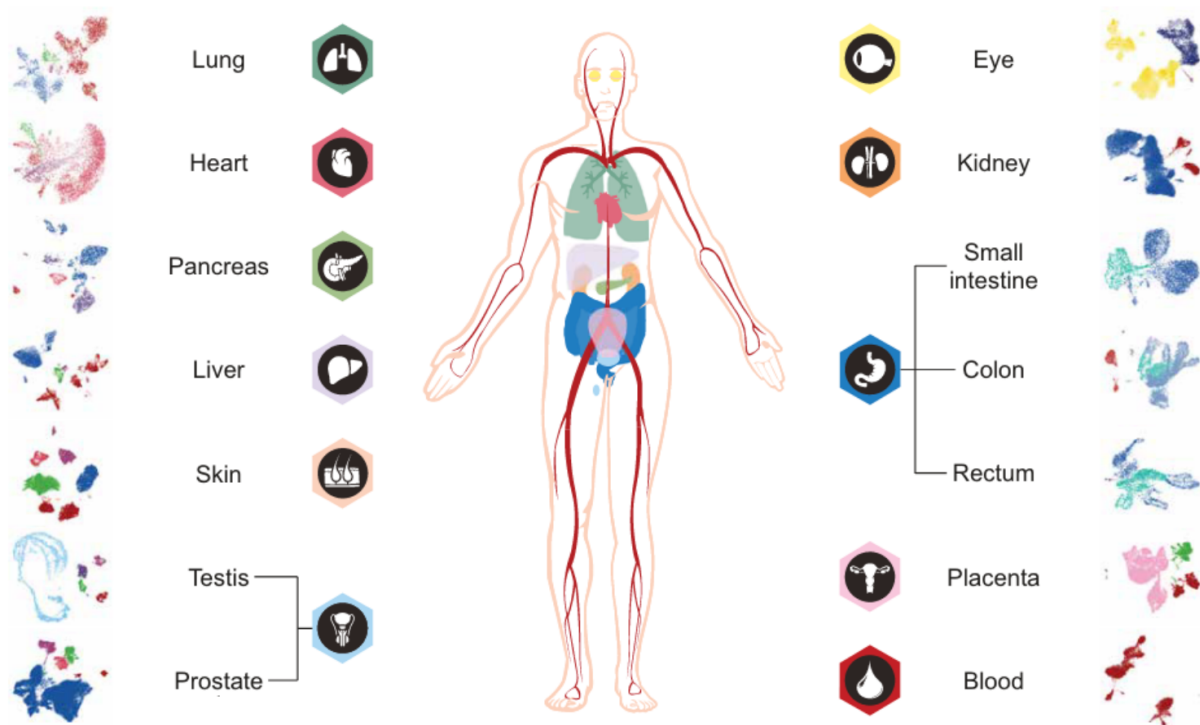


Figure 1.1: Diverse cell types across various tissues

Source: Karlsson, M., Zhang, C., Méar, L., Zhong, W., Digre, A., Katona, B., ... & Lindskog, C. (2021). A single-cell type transcriptomics map of human tissues. *Science advances*, 7(31), eabh2169. <http://doi.org/10.1126/sciadv.abh2169>

A cell takes part in plenty of biological scenarios at the same time. To gain a deeper understanding of cells, we can analyze the cell identity from various aspects. Firstly, we can segregate cells into diverse categories, which potentially can be broken down further into subcategories. Additionally, Different locations may have diverse cell-type compositions. Furthermore, we can recognize a cell at a distinct stage of its growth trajectory or during a particular phase of the cell cycle. Lastly, the cell can be distinguished based on its spatial

location, which shapes its external surroundings, the way it interacts with adjacent cells, and its developmental signals (Wagner et al., 2016).

As depicted in Figure 1.2, The blue cell undergoes various circumstances that jointly determine its identity. One of the primary factors influencing biological processes is environmental stimuli, which might include the accessibility of nutrients or the interaction of a signalling molecule with a receptor. At a particular stage in the developmental trajectory, the cell cycle and a spatial context are two distinct perspectives of cell type identification (Wagner et al., 2016). Within the spatial context, spatial domain detection tasks can be performed, identifying distinct spatial regions in the biological tissue section.

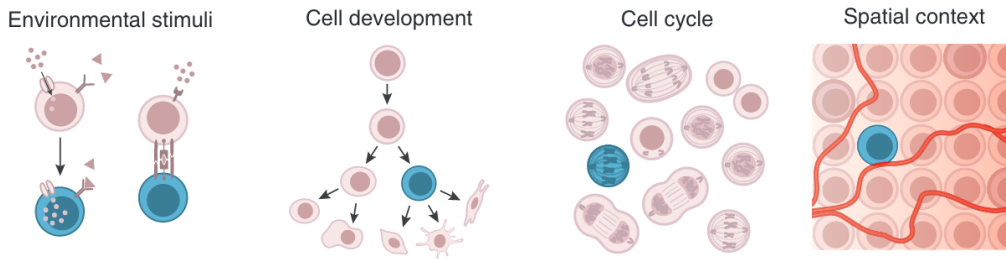


Figure 1.2: Different biological contexts of cell identification

Source: Wagner, A., Regev, A., & Yosef, N. (2016). Revealing the vectors of cellular identity with single-cell genomics. *Nature biotechnology*, 34(11), 1145-1160. <https://doi.org/10.1038/nbt.3711>

In the last few years, the emergence of single-cell genomics technology has provided a new stimulus for our increasing understanding of cellular identification and state.

1.2 Single-cell technologies and data structures

1.2.1 Data structures comparison

There are three broad classes of technologies that investigate the transcriptomics signal at individual resolutions. Figure 1.3 provides a description of the corresponding data structure. For scRNA-seq, which measures expression on dissociated cells, the data structure is typically shown as a cell-by-gene expression matrix. Regarding 2D spatially resolved transcriptomics technology, the data structure is presented as a spot-by-genes expression matrix, where each spot has a different cell type composition and possesses its own spatial x and y coordinates. Furthermore, 3D spatially resolved transcriptomics advances this concept by jointly aligning multiple slices in 2D spatially resolved transcriptomics and stacking them together to introduce the Z coordinate and facilitate 3D reconstruction.

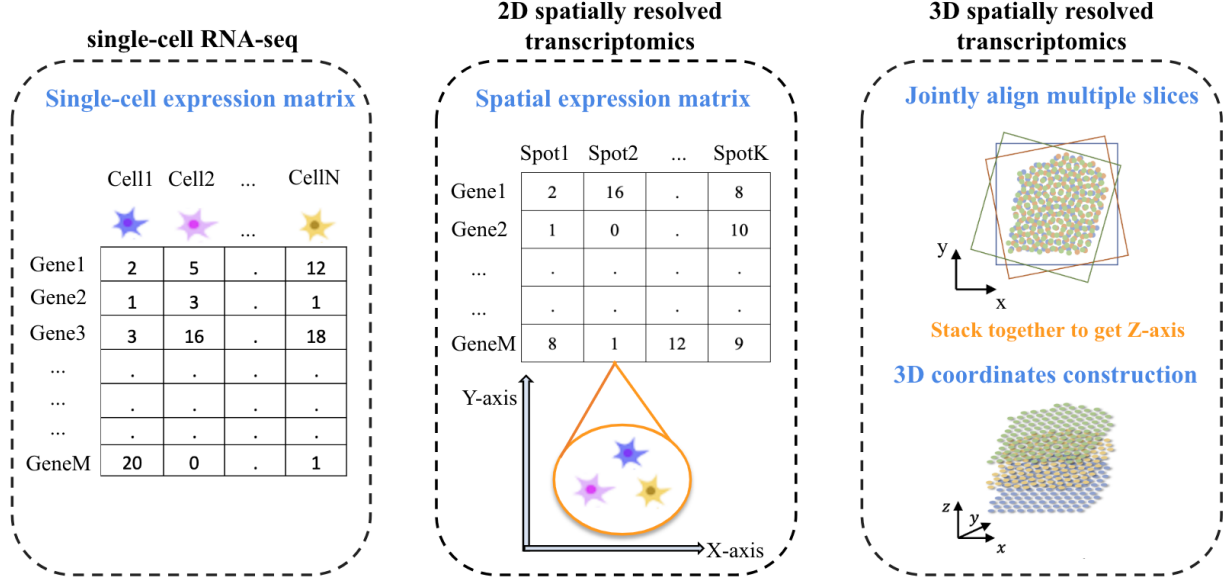


Figure 1.3: Single cell technologies data structure comparison

1.2.2 Single-cell RNA sequencing technology

Due to breakthroughs in advancements in microfluidics and RNA isolation and amplification techniques, we can now profile the transcriptome at individual cell's level by utilising the next-generation sequencing technology (Kiselev et al., 2019). Subsequently, technological advancements progressed quickly, with the first single-cell RNA sequencing (scRNA-seq) experiment being launched in 2009 (Tang et al., 2009), albeit limited to just eight cells. 10X Genomics provided single-cell gene expression datasets later in 2017. The abundance of scRNA-seq data has allowed us to provide a thorough classification of the cells for one specific sample (Kiselev et al., 2019). Nowadays, the scRNA-seq dataset for analysis is very large, even the transcriptome of approximately two million individual cells have been profiled (Cao et al., 2019).

Currently, researchers have the ability to deeply analyse the gene expression profile of individual cells with the help of the scRNA-seq technique (L Lun et al., 2016). The effective computational analysis of single-cell data is crucial in fully utilising these abundant datasets. In order to get a valuable biological result from the analysis, this involves processes including data quality control, read mapping, normalisation, dimensionality reduction, clustering, identifying heterogeneous cell populations, assessing differential gene expression, and trajectory inference (Wang et al., 2023b).

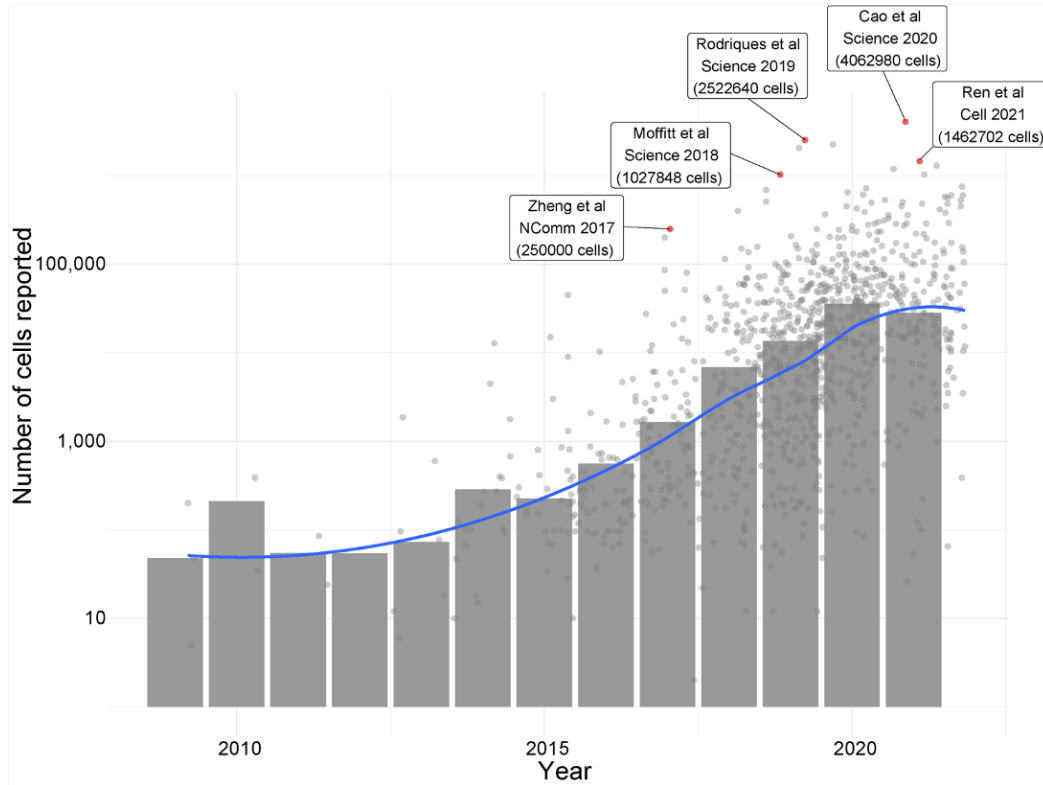


Figure 1.4: Number of cells reported per study since 2009

Source: Lin, Y. (2022). Statistical modelling and machine learning for single cell data harmonisation and analysis (Doctoral dissertation, The University of Sydney). School of Mathematics and Statistics. <https://hdl.handle.net/2123/28034>

With the rapid development of single-cell technologies, the scope of single-cell studies has expanded dramatically in recent years (Lin, 2022). As depicted in Figure 1.4, the median count of cells per study was 1,642 in 2017, which increased to 28,295 in 2021 (Svensson et al., 2020). The most extensive single-cell study to date, published in 2020, demonstrated a human fetal gene expression atlas covering an astounding 4,062,980 cells from 15 fetal organ samples (Cao et al., 2020).

1.2.3 Two-dimensional spatially resolved transcriptomics

In light of the persistent progress and milestones in scRNA-seq technology, a notable technological challenge still stands in our way. This challenge pertains to the requirement of effectively isolating viable cells from intact tissue samples while minimizing the detrimental effects of stress, cell demise, and cell aggregation (Williams et al., 2022). The field of immunology has experienced significant advancements through the utilization of scRNA-seq data (Haque et al., 2017). However, other cell types, like neurons in the brain, are more challenging to analyze using scRNA-seq data due to the need for specialized tissue dissociation techniques for their recovery (Lacar et al., 2016; Maynard et al., 2021).

Therefore, there exists a motivation to do transcriptomics analysis on entire tissue from the perspective of technology. As investigating whole tissue retains spatial information, these techniques are often called spatially-resolved transcriptomics or spatial transcrip-

tomics (ST) (Williams et al., 2022). To thoroughly understand the molecular intricacies of tissues, this technique intends to identify gene expression patterns while preserving the spatial tissue context information (Burgess, 2019).

The spatial location of a specific cell, in relation to adjacent cells and non-cellular structures, can provide valuable insights into characterizing the cellular phenotype, cell state, as well as the function of the cell and entire tissue. The position of cells is critical in analyzing biological signals since the exposure of cells to signals can be influenced by their location (Williams et al., 2022). For instance, breast cancer Cancer-associated fibroblasts (CAFs) can determine cell types based on expression and location (Cords et al., 2023). Recent literature suggests two methods to analyze transcriptomes while maintaining spatial information (Waylen et al., 2020; Moses and Pachter, 2022): Imaging-based spatial transcriptomics technologies and next-generation sequencing (NGS) methods for resolving gene expression spatially.

In this thesis, we mainly focus on the sequencing data. Sequencing technologies are used for the purpose of capturing messenger RNAs (mRNAs) from the tissue. Subsequently, complementary DNAs (cDNAs) are synthesized, and the abundance of gene-specific sequences is quantified using NGS methods. Also noteworthy is that spatial information is preserved at capturing mRNA (Williams et al., 2022). In the year 2016, spatial transcriptomics (ST) technology was capable of collecting positionally-barcoded mRNA using an ‘array-based’ approach (Ståhl et al., 2016). This ST platform captures the observation spots with a diameter of $100\ \mu\text{m}$. Then, 10X Genomics developed an advanced Visium platform, which can capture $55\ \mu\text{m}$. Consequently, based on the specific biological tissue being analyzed, the count of cells in a spot might reach as many as 200 on the earlier ST platform while varying from 1 to 30 on the Visium platform (Zhao et al., 2021). Figure 1.5 shows the difference between ST and Visium technologies, indicating that the ST platform is square-shaped, while the Visium platform is hexagonal.

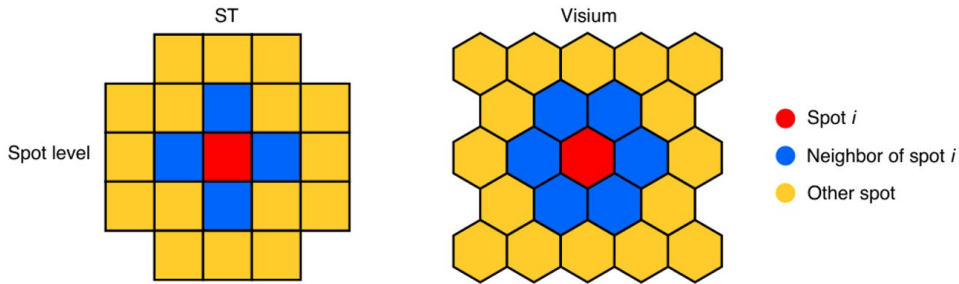


Figure 1.5: Geometric representations of spatial distribution of spots in the ST and Visium technologies

Note: For each individual spot (red), they have several neighboring spots (blue) based on common edges. Reference: Zhao, E., Stone, M. R., Ren, X., Guenthoer, J., Smythe, K. S., Pulliam, T., ... & Gottardo, R. (2021). Spatial transcriptomics at subspot resolution with BayesSpace. *Nature biotechnology*, 39(11), 1375-1384.

<https://doi.org/10.1038/s41587-021-00935-2>

Moreover, there are various platforms in spatial transcriptomics, with recent innovations including sequencing-based methods like Slide-seq (Rodriques et al., 2019) and fluorescence in situ hybridization (FISH) technologies such as seqFISH (Shah et al., 2016). Even though these approaches can enhance resolution, many of them come with drawbacks such as lower throughput, decreased sensitivity, reliance on customized protocols, or limited availability (Zhao et al., 2021).

Figure 1.6 illustrates a typical analysis workflow for sequencing data. Firstly, sequencing data undergoes preprocessing via alignment and mapping techniques. Subsequently, we proceed with data preprocessing. Within this process, one crucial step is the implementation of normalization procedures to remove unwanted technological variability. Following this, particularly with spot-based data, an intermediate analysis stage is often required. During this stage, deconvolution techniques can determine the cell-type composition within different spots. After obtaining the cell-type composition, clustering methods can be applied to the spatial domain detection task. For broader downstream analysis, packages such as Seurat (Satija et al., 2015), Scanpy (Wolf et al., 2018), Giotto (Dries et al., 2021), and stLearn (Pham et al., 2020) can be implemented. At the end of this process, biological inferences can be derived from the analyzed data, subsequently generating biologically significant insights and interpretations.

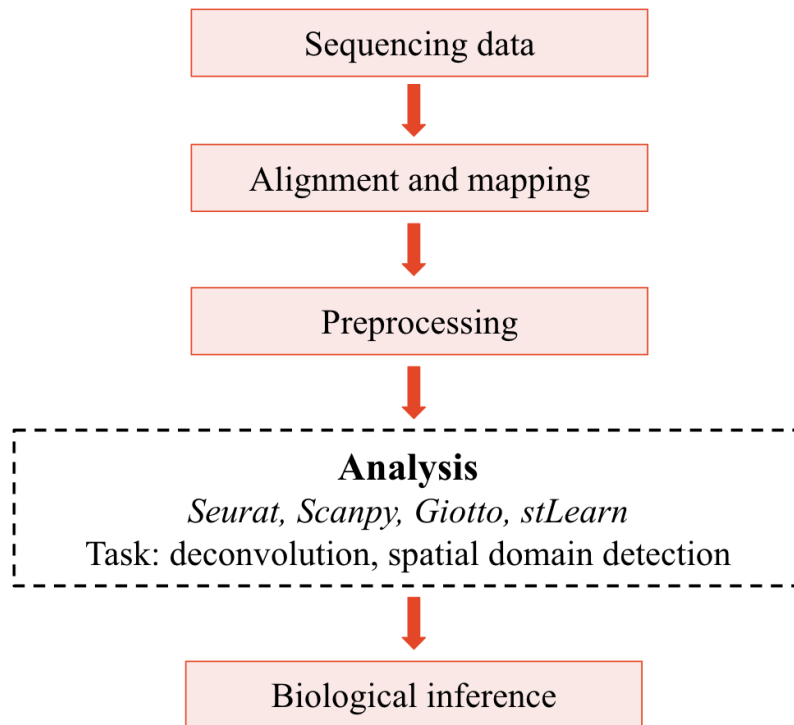


Figure 1.6: Sequencing data workflow

1.2.4 Three-dimensional spatially resolved transcriptomics

At present, the analysis of ST data is predominantly confined to 2D dimensions in a single tissue section, which constrains our ability to fully comprehend real biological signals occurring in a three-dimensional (3D) environment (Wang et al., 2023a). There is a growing proliferation of ST datasets that consist of multiple parallel two-dimensional (2D) slices in a specific tissue, oriented with the x and y-axis along the z-axis (Maynard et al., 2021; Asp et al., 2019; Wang et al., 2022). However, most current ST methods offer analysis outcomes for just one specific 2D tissue slice. Importantly, biological organs and embryos exhibit intricate 3D structures, and biological activities are seldom confined in a singular 2D space. The limited perspective provided by a 2D representation of a single tissue slice greatly constrains our ability to comprehend and analyze biological signalling and processes (Wang et al., 2023a). The simultaneous analysis of multiple 2D tissue sections provides potential opportunities to depict a holistic 3D depiction of biological systems to discover valuable biological revelations (Li and Zhou, 2022).

Nonetheless, the joint modelling of multiple spatial slices introduces challenges due to spatial warping, coordinate system, spatial resolution, and biological variance among different slices, samples, and individuals (Jones et al., 2022). Recently, there have been a number of computational methods that begin to provide solutions to address these challenges. These developed methods can model multiple parallel 2D tissue slices based on attaining spot alignment across adjacent tissue sections, such as the Probabilistic Alignment of Spatial Transcriptomics Experiments (PASTE) (Zeira et al., 2022) and Gaussian Process Spatial Alignment (GPSA) (Jones et al., 2022) algorithms.

The first PASTE algorithm utilizes an optimal transport formulation to calculate pairwise alignments of slices. The approach takes into account both physical distances and transcriptional similarity between locations. Furthermore, PASTE uses pairwise alignments to construct a 3D alignment by stacking them together (Zeira et al., 2022). Figure 1.7 depicts the workflow of the PASTE algorithm, highlighting that the purpose of the PASTE method is to identify gene expression patterns within their natural 3D tissue context. Additionally, it provides us with the opportunity to delve into the intricacies of tissue structure and conduct downstream analysis, such as clustering (Lin and Yang, 2022).

Additionally, the GPSA algorithm is a Bayesian model that facilitates the spatial alignment and analysis of numerous genomes and histological slices with spatial resolution. This model employs a combination of two Gaussian processes (GPs) arranged in a stacked manner to align multiple slices across various technologies and samples in a spatial context effectively. This alignment is achieved within a coordinate system that can be either 2D, 3D, or potentially even spatiotemporal in nature (Jones et al., 2022).

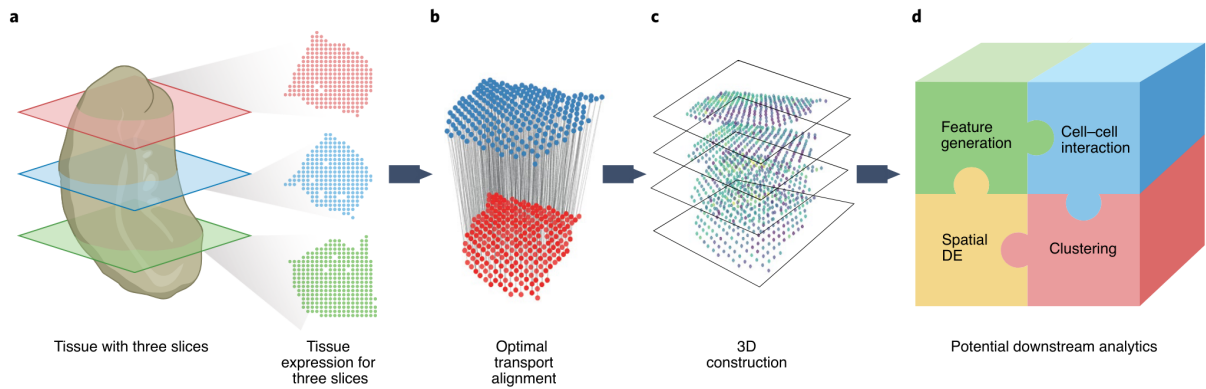


Figure 1.7: Aligning and integrating spatial transcriptomics data across adjacent slices

Source: Lin, Y., & Yang, J. Y. (2022). 3D reconstruction of spatial expression. *Nature Methods*, 19(5), 526-527.

<https://doi.org/10.1038/s41592-022-01476-5>

Joint modelling of multiple tissue slices is an initial step for us to perform additional 3D spatial domain detection. Following that, we can also perform additional 3D investigations to explore the fine-grained 3D structures within intact tissue structures. It allows us to discover genes exhibiting 3D spatial layouts and to discern gradients of 3D cell types within spatial tissue structures (Wang et al., 2023a). To better understand the comprehensive biological structure in 3D space, several advanced technologies have been developed, enabling us to perform spatial domain detection tasks in a 3D context.

1.3 Data Science challenges

1.3.1 Common challenges in single-cell RNA-seq data

The scRNA-seq data introduces a variety of inherent challenges, such as extreme sparsity, technical noise, and the complicated nature of the data (Yuan et al., 2017).

The first challenge is extreme sparsity, where the single-cell gene expression matrix contains a substantial percentage of zero values, stemming from biological and technical elements (L Lun et al., 2016). This zero count for a specific feature in a particular cell means the feature does not exist or there is not enough read coverage to detect it, a phenomenon often referred to as “dropout” (DeBruine et al., 2021). Due to heterogeneity across cells and transcriptional bursting events, there is a significant variation in gene expression across different cells (Marinov et al., 2014). This variability can lead to zero counts for genes expressed at low levels. The conversion of small amounts of input RNA into sequenceable libraries is also difficult to accomplish in technical ways (L Lun et al., 2016). Since small amounts of transcripts are missed during the preparation of the library, there are plenty of dropouts (Brennecke et al., 2013).

The zeros can be divided into three types: systematic, semi-systematic, and stochastic zeroes (L Lun et al., 2016). The first is systematic zeros, which mean genes that are

consistently silent throughout the entire dataset’s cells, suggesting a zero count for every cell. These often do not cause problems because they do not contain any information and could be eliminated before normalization. Semi-systematic zeros are slightly different from systematic zeros. They characterize genes that are silent in certain cell populations but actively expressed in others. This results in zero counts for those subgroups but still offers insights into the disparities between different subgroups. The last type is stochastic zeros, which appear in active gene expression, but there are some stochastic variations in sampling, resulting in zero counts. Given that these genes can provide important information, the dataset may be biased if we discard them before the normalization process (L Lun et al., 2016).

In addition, the scRNA-seq experiment presents a huge amount of technical noise. Compared to bulk RNA-seq, gene expression analysis using scRNA-seq naturally generates more noise because sparse sampling and substantial amplification of small quantities of initial material result in notable distortions (Yuan et al., 2017).

Furthermore, the scRNA-seq analysis strongly depends on the precise similarity task for a pair of cells, which presents unique difficulties such as outlier cell populations, noise from transcript amplification, and dropout events (Xiang et al., 2021).

1.3.2 Unique challenges in spatial transcriptomics

The spatial transcriptomics technology presents unique challenges. Firstly, while sequencing-based spatial transcriptomics techniques possess fine resolution and genome-wide transcript coverage, they have some drawbacks in transcript capture depth, leading to information loss for genes expressed at relatively lower expression levels (Choe et al., 2023). Moreover, another challenge arises from the non-cellular resolution of spatially resolved transcriptomics data, where a single spot contains a mixture of multiple cells, exhibiting the lower resolution characteristic of Spatial Transcriptomics techniques (Williams et al., 2022). This challenge also poses obstacles for deconvolution methods, which aim to identify the cell-type composition within each mixed spot (Li et al., 2023).

1.4 The aim of the thesis

The realm of spatial clustering in spatially resolved transcriptomics data remains relatively unexplored, with limited studies providing a comprehensive evaluation of clustering methods. To fill this critical gap, we propose two research questions to assess the performance of clustering methods in spatially resolved transcriptomics data. The first question explores how clustering methods utilise spatial coordinates (X-axis and Y-axis) compared to classical clustering approaches. The second question aims to investigate whether using 3D coordinates in clustering methods improves spatial domain detection performance com-

pared to methods relying only on 2D coordinates. By addressing these two questions, this thesis aims to provide valuable insights into the highlighted gap in the current research on spatial clustering in spatially resolved transcriptomics data.

1.5 Outline of thesis

In Chapter 2, we will conduct a comprehensive literature review on clustering analysis within single-cell fields and describe the various clustering methodologies we plan to investigate in the following chapter. Chapter 3 will compare classical clustering methods with 2D spatial clustering methods in the context of two-dimensional spatial domain detection. Following this, Chapter 4 will carry out a comparison between 2D and 3D spatial clustering methods in spatial domain detection. We will conclude this thesis in Chapter 5 by outlining some limitations of the present study and suggesting potential directions for future research.

Chapter 2

Literature review on clustering analysis in single-cell field

In this chapter, we perform a comprehensive literature review on clustering algorithms in the single cell, including classical clustering methods for scRNA-seq, 2D spatial clustering methods, and 3D spatial clustering methods. Before discussing the algorithms for different clustering methods, we first define some parameters that will be used in the subsequent method descriptions.

- **Cell type labels:** The number of distinct cell types is represented by g , and G represents the total cell types identified in the dataset.
- **Spatial domain labels:** Different spatial domain labels are represented by $C = \{c_i = 1, \dots, k\}$.
- **Spatial coordinates:** The coordinates are given by x , y , and z for X -axis, Y -axis, and Z -axis, respectively.

2.1 Single-cell RNA sequencing clustering analysis

One of the key questions of interest in the single-cell field, scRNA-seq data has become larger and more diverse with the new incorporation of spatial information, driving the need for classification methods that can provide more refined discrimination between cell type and cell states. There are two broad classes of cell-type identification approaches. Firstly, unsupervised learning or clustering, which does not depend on pre-train data and is commonly believed to enable novel cell type identification. Secondly, supervised learning, where we leverage much-existing information. In this section, we focus on unsupervised learning or clustering to address the question of cell type identification.

The procedure preceding the clustering algorithm can have a significant impact on the clustering results. Many methods have been developed and can be implemented at different stages. While many integrated software packages are available to perform the clustering process, it is critical to select the right tools and methods for a particular dataset and research objective.

As depicted in Figure 2.1, quality control, normalization, feature selection, dimension reduction, and detecting cell-cell distances are necessary pre-processing steps for imple-

menting unsupervised clustering on scRNA-seq data. The spatial transcriptomics technology presents unique challenges. Firstly, while sequencing-based spatial transcriptomics techniques possess fine resolution and genome-wide transcript coverage, they have some drawbacks in transcript capture depth, leading to information loss for genes expressed at relatively lower expression levels (Choe et al., 2023). Moreover, another challenge arises from the non-cellular resolution of spatially resolved transcriptomics data, where a single spot contains a mixture of multiple cells, exhibiting the lower resolution characteristic of Spatial Transcriptomics techniques (Williams et al., 2022). This challenge also poses obstacles for deconvolution methods, which aim to identify the cell-type composition within each mixed spot (Li et al., 2023).

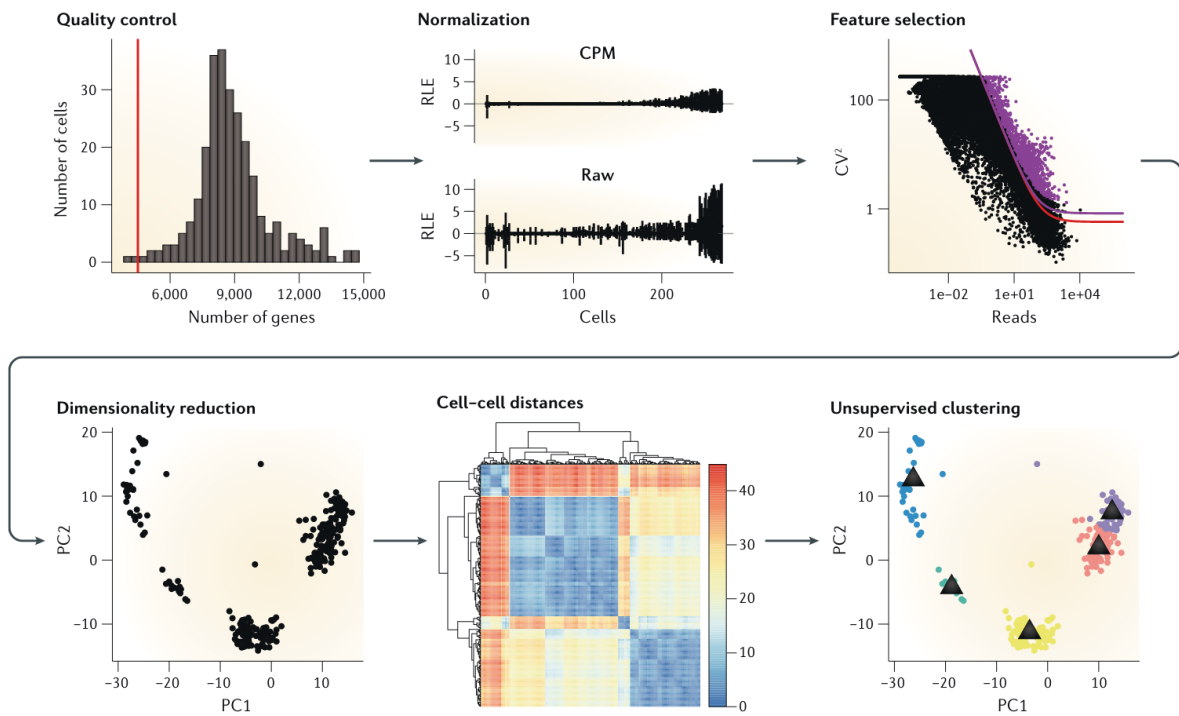


Figure 2.1: Unsupervised clustering workflow for scRNA-seq data. CPM, counts per million; CV, coefficient of variation; PC, principal component; RLE, relative log expression.

Source: Kiselev, V. Y., Andrews, T. S., & Hemberg, M. (2019). Challenges in unsupervised clustering of single-cell RNA-seq data. *Nature Reviews Genetics*, 20(5), 273-282. <https://doi.org/10.1038/s41576-018-0088-9>

To be more specific, we will go into further detail regarding the steps required before applying unsupervised clustering to classify different cell types:

- **Quality control:** Implementing quality control prior to the downstream analysis is essential due to the substantial technical noise in scRNA-seq data. The Scater R package is one of the most widely used for performing quality control. On the one hand, quality control involves the removal of low-quality cells to prevent distortion arising from technical noise. On the other hand, in order to boost computational

efficiency without losing valuable information, it is also essential to discard genes that are either low abundance or exhibit high dropout rates (McCarthy et al., 2017).

- **Normalization:** After quality control, the data set is subsequently normalized to adjust for disparities in read coverage and other technical confounding factors (Kiselev et al., 2019). The most original method is to apply log normalization transformation to raw scRNA-seq data to ensure all data falls within a specific range. Additionally, there exists plenty of advanced normalization techniques to reduce noise or bias specifically for scRNA-seq data, such as the R package *scran* (L Lun et al., 2016), *SCnorm* (Bacher et al., 2017) and *Linnorm* (Yip et al., 2017).

1. *Scran* is a deconvolution method that pools cells together and establishes a linear system depending on the total expression in each pool of cells (L Lun et al., 2016).

2. *SCnorm* utilizes quantile regression to measure the relationship between transcript expression and the sequencing depth for each gene. Following this, genes exhibiting similar dependencies are clustered together, followed by the application of second quantile regression to assess scale factors within each cluster (Bacher et al., 2017).

3. *Linnorm* technique is a normalizing transformation method grounded in the ideas of linear models and normality principles, which is analogous to the scaling factor. The advantage of this approach is its ability to eliminate technical noises while retaining biological variations present in scRNA-seq data (Yip et al., 2017).

- **Feature selection:** Identifying a collection of important genes, commonly known as ‘features’, is an essential stage. These expression patterns are in a lower-dimensional space, which will be used for the following clustering analysis. The features capable of differentiating between various cell types are deemed a nice subset, and the quality of this collection can hugely affect the precision of the clustering (Su et al., 2021).
- **Dimensionality reduction:** Dimensionality reduction plays a crucial role in the downstream analysis given that single-cell datasets exhibit high dimensionality owing to a vast number of measurement cells. In addition, biological systems inherently possess a lower dimensionality. Hence, dimensionality reduction is essential for converting datasets with high-dimensional into low-dimensional space, enabling the visualization of clustering outcomes and the deduction of developmental trajectories (Xiang et al., 2021).
- **Cell-cell distances:** Following the reduction of dimensions, distances between cells are computed in the lower-dimensional space. These distances can be utilized to create a cell-cell distance graph or can be directly employed by clustering algorithms

to classify cells into specific categories. Some approaches, however, calculate these distances prior to implementing the dimensionality reduction (Kiselev et al., 2019).

After completing the above pre-processing steps, we can use clustering algorithms to identify cell types. The basic definition of the unsupervised clustering algorithm is to classify items according to similarity rather than utilizing training data with labels or ground truth. Transcriptome-based cell type identification is appealing since it offers a data-driven, consistent, and impartial way, which is adaptable to every sample (Kiselev et al., 2019). For understanding cell biology, clustering algorithms play a pivotal role in assisting biologists in comprehending how biological processes work. These computational tools help categorize cells based on their similar expression patterns, giving biologists a deeper insight into the inherent structures and functions of various cell types.

2.2 Clustering methods for scRNA-seq data

There are now a variety of clustering techniques for scRNA-seq data. We will categorize and discuss this clustering algorithm into three main groups, namely K-means clustering, hierarchical clustering, and community connection.

2.2.1 K-means based clustering methods

The most well-known clustering algorithm is K-means. It assigns each cell to the nearest cluster centre following recurrent identification of the k cluster centres, also known as centroids (Kiselev et al., 2019). The conventional approach for K-means is Lloyd’s algorithm, which possesses the benefit of exhibiting linear scalability with respect to the set of points, enabling its application to substantial datasets like scRNA-seq data. However, this algorithm cannot ensure the identification of the optimal solution (Lloyd, 1982). These limitations can be addressed by utilizing the SC3 (single-cell consensus clustering) technique, which implements the K-means algorithm repeatedly under various initial settings or upstream processing, eventually determining a consensus (Kiselev et al., 2017).

In addition, the K-means algorithm has another drawback, which is its tendency to favour the identification of clusters of identical size, potentially leading to uncommon cell types being obscured in the bigger clusters (Kiselev et al., 2019). To address these problems, the RaceID (rare cell type identification) method applies the K-means algorithm integrated with outlier identification, detecting outliers that are unaccounted for by a background model considering both gene expression noise, making it easier to find uncommon cell types (Grün et al., 2015). Moreover, another technique known as SIMLR (single-cell interpretation through multi-kernel learning) is also capable of addressing this issue, which adjusts K-means by developing a customized distance metric at the same time (Wang et al., 2017).

2.2.2 Hierarchical based clustering methods

Another widely used clustering algorithm for scRNA-seq is hierarchical clustering. There are two categories for the hierarchical clustering. The first method is agglomerative hierarchical clustering, which involves integrating single cells into bigger groups in a step-by-step manner. The second strategy, known as divisive hierarchical clustering, partitions groups into smaller subsets. One notable limitation of hierarchical clustering algorithms is that they are very computationally expensive when using large scRNA-seq datasets, primarily due to significant demands on time and memory resources (Kiselev et al., 2019). The CIDR (Clustering through Imputation and Dimensionality Reduction) technique modifies hierarchical clustering for scRNA-seq by incorporating dropouts into the distance computation. With this improvement, cell-to-cell distance measurements in samples with low sequencing depth are more reliable (Lin et al., 2017).

Furthermore, by performing dimensionality reduction after the merging or splitting in every step, several methods rely on the algorithms of hierarchical clustering (Kiselev et al., 2019). This recurring approach enhances the capacity to identify small classifications, applying in BackSPIN (Zeisel et al., 2015) and pcaReduce (Žurauskienė and Yau, 2016) technique. BackSPIN technique utilizes a divisive biclustering approach grounded on the concept of Sorting Points Into Neighborhoods (SPIN) (Zeisel et al., 2015), whereas the pcaReduce method employs an alternative agglomerative clustering method that incorporates principal components analysis (PCA) and hierarchical clustering (Žurauskienė and Yau, 2016).

2.2.3 Community detection based clustering methods

Given the many shortcomings of both K-means and hierarchical clustering algorithms, especially when dealing with large scRNA-seq data, the famous community detection clustering algorithm based on graphs can be taken into consideration. The community detection methods determine clusters of nodes that display dense connectivity rather than detecting sets of closely located points. To begin with, we need to construct the k -nearest-neighbours(KNN) graph. When building the single-cell graph, we should select a value for the k value, which denotes the nearest neighbours. The decision of k will influence the count and scale of the groups in the clustering results (Kiselev et al., 2019).

The advantage of using a community detection algorithm is that this technique utilizes indirect resolution parameters, so we do not need to predefine the number of clusters before the clustering. Besides, compared to hierarchical clustering, this method demonstrates more efficiency and has a shorter running time (Kiselev et al., 2019). Louvain community detection (De Meo et al., 2011) approach is one of the most popular methods in community detection, which is commonly used in single-cell analysis. However, Traag et al. have demonstrated that the Louvain algorithm has a drawback: it may sometimes produce

communities with poor connectivity. In order to solve this issue, they invented a new community detection method named Leiden, ensuring better community connectivity (Traag et al., 2019). In addition, the PhenoGraph (Levine et al., 2015) method is more advanced since it incorporates the shared-nearest-neighbour graph and Louvain algorithm, and this method has been integrated into Seurat (Satija et al., 2015) in R and scanpy (Wolf et al., 2018) in Python.

2.2.4 Summary clustering methods for scRNA-seq data

Table 2.1 summarizes the specific clustering algorithms for scRNA-seq data from the three major clustering categories we have discussed by comparing them based on their method types, strengths, and limitations.

Table 2.1: Clustering methods for scRNA-seq data

Name	Year	Method type	Strengths	Limitations
SC3	2017	PCA+K-means	High accuracy through consensus, provides estimation of k	High complexity, not scalable
RaceID	2015	K-means	Detects rare cell types, provides estimation of k	Performs poorly when there are no rare cell types
SIMLR	2017	Data-driven dimensionality reduction+K-mean	Concurrent training of the distance metric improves sensitivity in noisy data sets	Adjusting the distance metric to make cells fit the clusters may artificially inflate quality measures
CIDR	2017	PCA+hierarchical	Implicitly imputes dropouts when calculating distances	
BackSPIN	2015	Biclustering (hierarchical)	Multiple rounds of feature selection improve clustering resolution	Tends to over-partition the data
pcaReduce	2016	PCA+K-mean+hierarchical	Provides hierarchy of solutions	Very stochastic, does not provide a stable result
PhenoGraph	2015	PCA+graph-based	Very scalable	May not be accurate for small data sets
Seurat	2016			
Scanpy	2018			

Source: Kiselev, V. Y., Andrews, T. S., & Hemberg, M. (2019). Challenges in unsupervised clustering of single-cell RNA-seq data. *Nature Reviews Genetics*, 20(5), 273-282. <https://doi.org/10.1038/s41576-018-0088-9>

2.2.5 Challenges in scRNA-seq clustering analysis

Clustering is a crucial process in distinguishing the subpopulation structure of different cells in scRNA-seq data, but it has several challenges during this analysis (Petegrosso et al., 2020).

First of all, due to a range of cell-specific features, including the stages of the cell cycle or the size of the cell, technical noise and biases are present. Additionally, these technical noises and biases arise from technical or systematic factors such as inefficient capture processes, amplification biases, and the depths of sequencing (Petegrosso et al., 2020), which in turn cause irregular transcriptome coverage, leading to a substantial number of regions with zero coverage and a high prevalence of “dropout” genes (Hebenstreit, 2012; Bacher and Kendzierski, 2016).

Moreover, when examining different single-cell populations from a collection of samples as a whole, technical biases and biological variations tend to overshadow the clustering of individual cells in most scenarios. Consequently, the cells are more likely to cluster based on their origin in the sample rather than similar cell types (Zhang et al., 2018).

Additionally, there are other difficulties in the clustering algorithm arising from the “curse of dimensionality”, which refers to the issue in scRNA-seq, where a plethora of genes are examined, leading to high-dimensional data spaces. In these spaces, the distances between cells tend to be similar, adding complexity to the clustering analysis. Figure 2.2 demonstrates that the closer distances between clusters imply that clusters become less evident with 2000 genes than with 500 genes. For example, the 4-cell stage lacks as much separation. Therefore, the “curse of dimensionality” makes unsupervised clustering increasingly difficult (Kiselev et al., 2019).

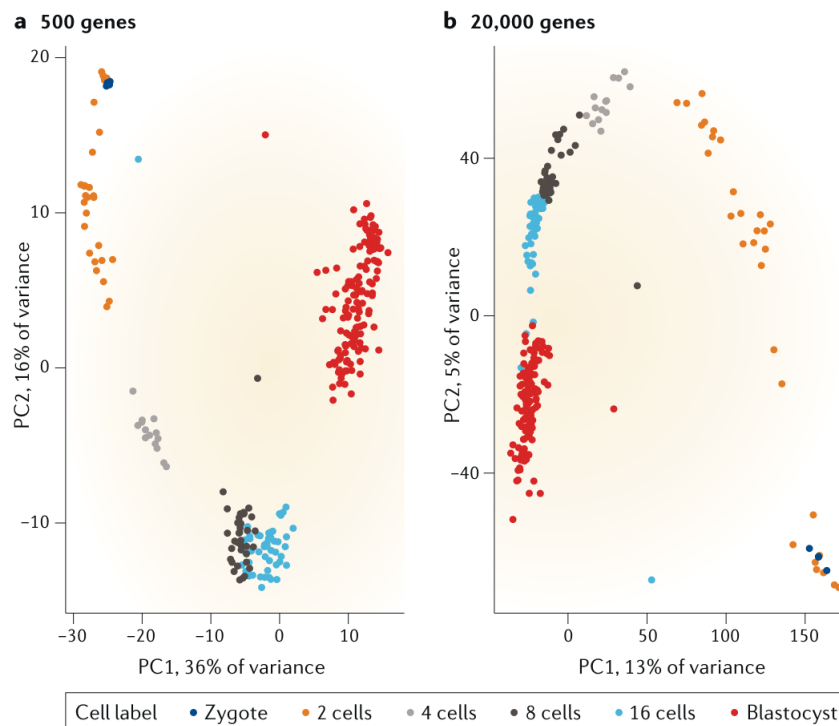


Figure 2.2: Illustration of the curse of dimensionality

Source: Kiselev, V. Y., Andrews, T. S., & Hemberg, M. (2019). Challenges in unsupervised clustering of single-cell RNA-seq data. *Nature Reviews Genetics*, 20(5), 273-282. <https://doi.org/10.1038/s41576-018-0088-9>

2.3 Spatial clustering methods for spatially resolved transcriptomics data

The primary goal of clustering scRNA-seq data is to determine cell types and divide cells into discrete categories based on their features. In contrast to scRNA-seq clustering, spatial clustering in spatial transcriptomics is designed to group spatial locations into separate spatial domains. This approach plays a critical role in comprehending tissue structure, enhancing our holistic understanding of the tissue’s spatial arrangement and functional organization. When investigating both scRNA-seq and spatial transcriptomics datasets, feature selection and dimension reduction play a vital role in implementing (spatial) clustering methods (Liu et al., 2022).

Feature selection: SVGs

The critical first step in the analysis of spatial transcriptomics data is detecting genes that exhibit spatial variation within spatial domains of tissues and cells, which are termed spatially variable genes (SVGs) (Chen et al., 2022). Identifying SVGs is crucial to recognize genes that possess biological signals and reduce the high-dimensional characteristics of spatial transcriptomics data (Svensson et al., 2018). Selecting spatial features for spatially resolved transcriptomics data is similar to identifying highly variable genes (HVGs) in conventional scRNA-seq data (Yip et al., 2019). However, there is a nuanced differentiation between the two: SVGs capture the spatial connections of tissues and cells, while HVGs may not accurately represent these connections (Chen et al., 2022). There has been a notable increase in the variety of techniques for the identification of SVGs, such as SpatialDE (Svensson et al., 2018), SPARK (Sun et al., 2020), SPARK-X (Zhu et al., 2021), nnSVG (Weber et al., 2023) and so on.

The spatialDE approach relies on Gaussian process regression, a widely used model in the field of geostatistics (Svensson et al., 2018). In addition, the SPARK technique employs generalized linear spatial models to represent spatial count data directly (Sun et al., 2020). However, these two approaches come with certain limitations, like unsuitable for processing extensive sparse spatially resolved transcriptomics data or not being optimal for constructing sparse count data models using Gaussian distribution.

To overcome the above obstacles, Zhu et al. introduced the SPARK-X method, a scalable non-parametric test designed for detecting spatial expression (SE) genes. Figure 2.3 depicts the schematic of the SPARK-X method for detecting SVGs. The objective of this method is to identify genes that exhibit spatial expression patterns, referred to as SE genes, in large spatially resolved transcriptomic data using a non-parametric test. Specifically, this approach first builds an expression covariance matrix from the gene expression levels, represented as an n by n matrix $E = y(y^T y)^{-1} y^T$. Subsequently, a matrix reflecting distance covariance for all samples is calculated using their spatial coordinates, denoted as an n by

n matrix $\Sigma = S(S^T S)^{-1} S^T$. These two matrices with symmetric and positive semi-definite properties are defined as covariance matrices since they are from the projection covariance function. Then we can calculate the test statistic: $T = \frac{\text{trace}(E\Sigma)}{n}$. Therefore, we can calculate a P value utilizing Davies' exact method to quantify the probability of observing the same or a more extreme T value in the data using the null distribution. Finally, we can use the P value to identify spatial expression patterns (Zhu et al., 2021).

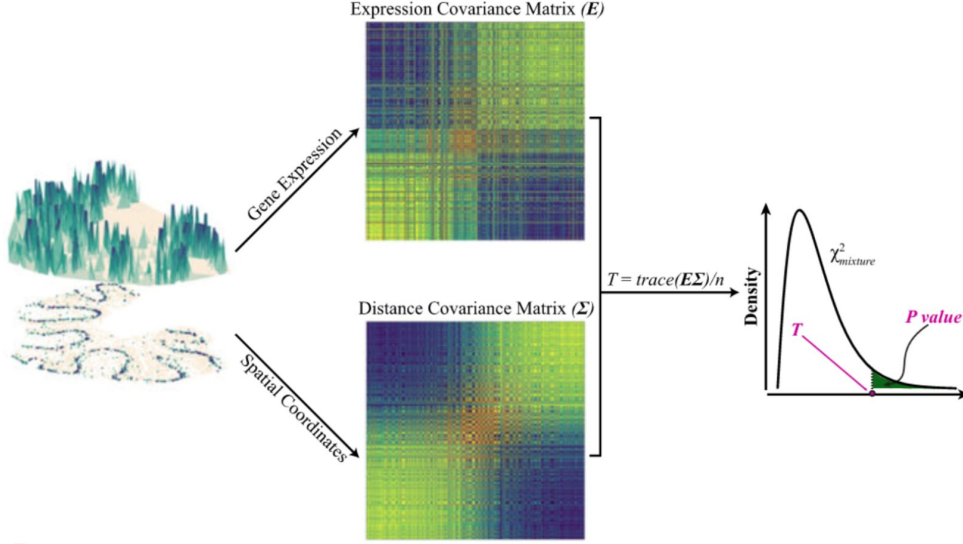


Figure 2.3: Method schematic of SPARK-X

Source: Zhu, J., Sun, S., & Zhou, X. (2021). SPARK-X: non-parametric modeling enables scalable and robust detection of spatial expression patterns for large spatial transcriptomic studies. *Genome biology*, 22(1), 1-25.
<https://doi.org/10.1186/s13059-021-02404-0>

Furthermore, the recently developed nnSVG method focuses on discerning SVGs based on nearest-neighbour Gaussian processes (Weber et al., 2023).

Dimension reduction: SpatialPCA

Dimension reduction is also an essential part of data processing, aimed at gathering meaningful biological signals from the noise present in spatially resolved transcriptomics data (Shang and Zhou, 2022). The application of dimension reduction techniques is widely used in scRNA-seq analyses (Sun et al., 2019). The principle component analysis (PCA) is recognized as a classical and widely used technique for dimension reduction in the single-cell field. Several dimensionality reduction methods have been specially tailored for scRNA-seq data, such as Zero Inflated Factor Analysis (ZIFA) (Pierson and Yau, 2015), specifically designed to take dropout events into account, and several of these methods have been directly utilized for the analysis of spatially resolved transcriptomics data.

However, these scRNA-seq dimension reduction approaches are not specifically designed for spatially resolved transcriptomics data, and they may not efficiently use spatial location information and spatial correlation patterns in tissue structures. In 2022, Shang and Zhou

presented a dimension reduction method with spatial awareness, termed spatial probabilistic PCA (SpatialPCA). One noteworthy property of SpatialPCA is its capacity to capture the spatial correlation patterns within tissue locations directly, ensuring the preservation of the raw data’s neighbouring similarities in the reduced-dimensional space (Shang and Zhou, 2022).

As shown in Figure 2.4, it illustrates the procedural diagram of the SpatialPCA dimension reduction technique. SpatialPCA is based on the probabilistic version of PCA, integrating spatial location information as an additional input and employing a kernel matrix to directly capture spatial correlation patterns within tissue positions. The components obtained by SpatialPCA in low dimensions are defined by their meaningful spatial correlation data. Therefore, these components can be referred to as spatial principal components (spatial PCs). Similar to using principal components (PCs) in downstream analysis of scRNA-seq data, these spatial PCs can be input for efficient downstream analysis, such as spatial domain detection in spatially resolved transcriptomics data (Shang and Zhou, 2022).

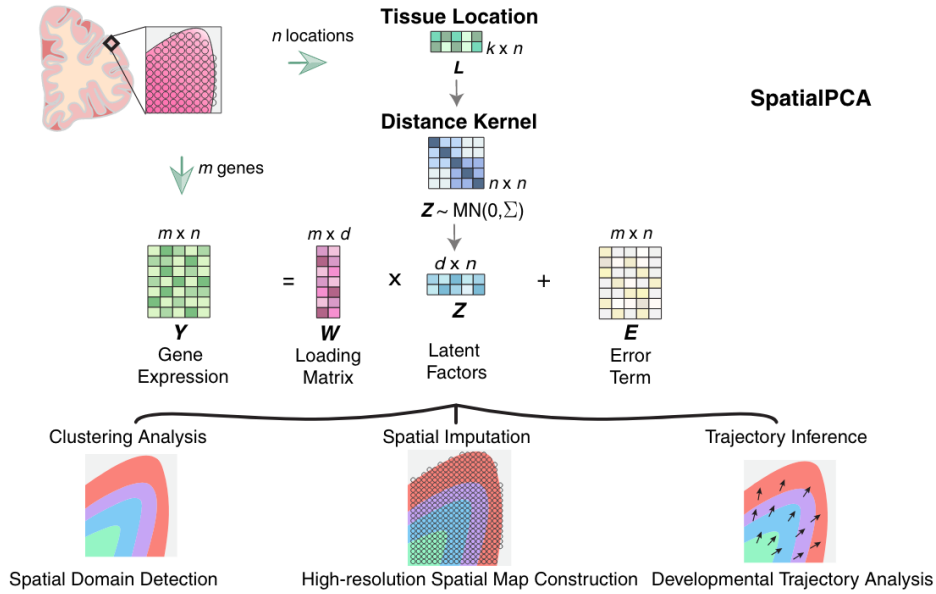


Figure 2.4: Method schematic of SpatialPCA

Source: Shang, L., & Zhou, X. (2022). Spatially aware dimension reduction for spatial transcriptomics. Nature Communications, 13(1), 7203. <https://doi.org/10.1038/s41467-022-34879-1>

After implementing the pre-processing steps using spatial tools, we can utilize 2D spatially resolved transcriptomics data on a single tissue slice (x, y) for spatial domain identification. To achieve a more in-depth understanding of the biological tissue structure, we can also perform 3D modelling (x, y, z) with spatially resolved transcriptomics by aligning multiple tissue slices and then detecting spatial domains. In the subsequent analysis, we will discuss the spatial domain detection techniques separately, focusing first on 2D and then on 3D.

2.3.1 2D spatial clustering methods

The utilisation of ST and Visium technologies can effectively tackle the analytical obstacles caused by sparsity and noise through the incorporation of extra spatial information, which is achieved by smoothing across neighbouring spots, which typically exhibit similar transcriptomic profiles (Zhao et al., 2021).

With the advancement of technology, many spatial clustering algorithms have been developed recently. To date, the spatial clustering methods can be broadly classified into four categories:

- (1) Markov random field (MRF) based spatial clustering
- (2) Deep learning based spatial clustering
- (3) Neighbourhood based spatial clustering
- (4) Integrated spatial clustering

The first category is Markov Random Field (MRF)-based spatial clustering. The HMRF (Zhu et al., 2018) model was proposed by Zhu et al., which aims to simultaneously model gene expression and spatial neighbourhood structure to delineate distinct spatial domains. Additionally, BayesSpace (Zhao et al., 2021) is a fully Bayesian method with a Markov random field (MRF) through the utilisation of a low-dimensional representation of the matrix of gene expression and classifying adjacent spots into the same group by a spatial prior. However, certain inherent limitations led to the development of SC-MEB. For example, smoothness is a vital parameter in MRF-based techniques, mainly influencing the closeness of adjacent spots, but BayesSpace treats this parameter as a constant, which restricts its capacity to pick the most suitable value corresponding to a specific dataset. To tackle these issues, the Spatial Clustering using the Hidden Markov Random Field based on Empirical Bayes (SC-MEB) (Yang et al., 2022) model integrates and enhances both the HRMF and BayesSpace algorithms, leading to the formulation of an effective expectation-maximisation (EM) algorithm.

Moving to the second category, methods like stLearn (Pham et al., 2020) employ deep learning characteristics that are taken from histopathology images, along with neighbouring spots' expressions, in order to achieve spatial smoothing of the data. Further enriching this category is SpaGCN (Hu et al., 2021), a graph convolutional network (GCN) that incorporates three essential pieces of information: gene expression, spatial location and histology information.

Furthermore, the third category is neighbourhood-based spatial clustering, such as Building Aggregates with a Neighbourhood Kernel and Spatial Yardstick (BANKSY) (Singhal et al., 2022). This method can utilise the idea that a specific cell in a tissue can be better understood by considering not only its own transcriptome but also the transcriptome of its neighbouring environment. This approach aims to implement cell-type clustering and spatial domain detection tasks in a unified framework incorporating feature augmentation.

Lastly, the integrated spatial clustering represents the confluence of multiple techniques, exemplified by methods such as Dimension-Reduction Spatial-Clustering (DR-SC) (Liu et al., 2022). This technique is a powerful method for spatial clustering that combines dimension reduction and spatial clustering into an integrated framework. Specifically, this method utilises a latent hidden Markov random field model to improve the spatial smoothness of the identified spatial cluster boundaries.

In the subsequent chapter, we will evaluate the clustering performance using the HMRF (Zhu et al., 2018), BayesSpace (Zhao et al., 2021), and BANKSY (Singhal et al., 2022) techniques. Consequently, an in-depth description of these selected spatial clustering methods is presented in the following sections.

2.3.1.1 2D spatial clustering methods

HMRF

The Hidden Markov Random Field (HMRF) (Zhu et al., 2018) model is a frequently employed graph-based technique utilised for pattern recognition to analyse spatially resolved transcriptomics data. The HMRF model is utilised to characterise the spatial distribution of a signal in the usual setting. The spatial organisation is depicted as a collection of nodes positioned on a regular grid, with adjacent nodes interconnected. Since it needs to be inferred from other directly observable variables indirectly, the spatial pattern is “hidden”. The primary key assumption of this model is the Markov property, which assumes that spatial constraints can be simplified by focusing solely on the correlation between adjacent neighbouring nodes. In 2021, this HMRF method has been incorporated into the Giotto package (Dries et al., 2021). For detecting the spatial domain detection, we can utilise this package to implement the HMRF algorithm.

By expressing this algorithm mathematically, let us denote the nodes in the graph as $S = \{S_i\}$. The combination of these nodes and the adjacency defined by the local neighbourhood graph form the neighbourhood system $(S, \{N_i\})$, where N_i represents the set of neighbouring nodes directly connected to node S_i in the graph, indicating its local neighbourhood. Each node i is linked with observed signal values, denoted by x_i . Moreover, let $C = \{c_i = 1, \dots, k\}$ be the set that indicates the spatial domain labels. The joint probability that a node S_i is related to class c_i is given by the subsequent equation:

$$P(c_i|s_i, x_i, c_{N_i}) = \frac{1}{Z} P(x_i|c_i, s_i) P(c_i|s_i, c_{N_i})$$

where $P(x_i|c_i, s_i)$ measures the node S_i 's own gene expression effect, while $P(c_i|s_i, c_{N_i})$ measures the neighbouring cells configuration c_{N_i} effects, which can be determined by the Gibbs distribution:

$$P(c_i|s_i, c_{N_i}) = \frac{1}{Z} \exp(-\beta \sum_{s_j \in N_i} U(c_j, c_i))$$

where $U(c_i, c_j)$ is termed the energy function. The precise formulation of $U(c_i, c_j)$ depends on the particular application and sets the assumption about how adjacent nodes interact with each other. In the HMRF model, the special case of Pott's model is employed.

$$U(c_j, c_i) = -1 \text{ if } c_j = c_i \text{ and } 0 \text{ otherwise}$$

which suggests that the neighbouring cells' effects are additive. In essence, $P(c_i|s_i, c_{N_i})$ represents the cumulative energy as the sum of pairwise interaction energies with neighbouring nodes. The parameter β indicates the strength of these interactions. A schematic overview of the HMRF model is presented in Figure 2.5.

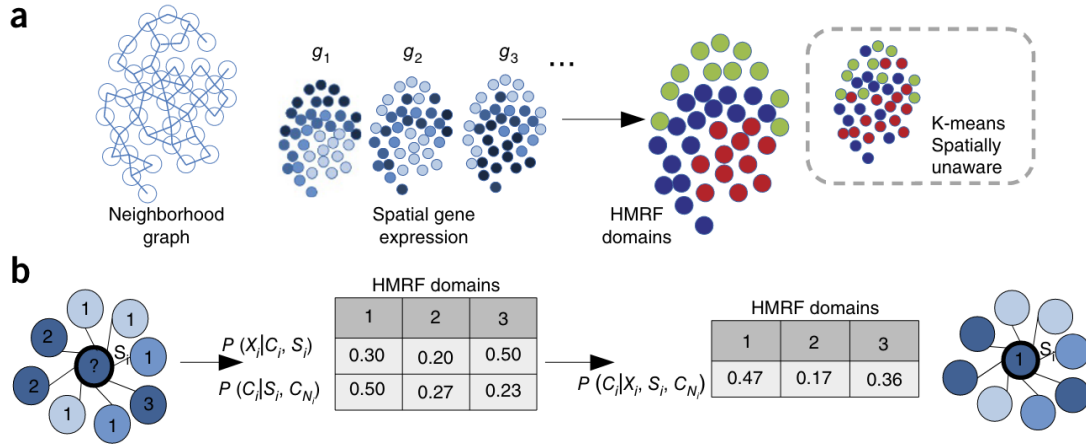


Figure 2.5: Schematic overview of HMRF

Source: Zhu, Q., Shah, S., Dries, R., Cai, L., & Yuan, G. C. (2018). Identification of spatially associated subpopulations by combining scRNAseq and sequential fluorescence in situ hybridization data. *Nature biotechnology*, 36(12), 1183-1190. <https://doi.org/10.1038/nbt.4260>

From (a) in this figure, the spatial relationship between cells (denoted by the circles), depicted via a neighbourhood graph, is represented by edges connecting adjacent cells. In order to detect spatial domains, the identified multigene expression profiles are utilized with the graph topology jointly. By contrast, K-means and other classical clustering techniques do not incorporate spatial information, so the results are anticipated to be less coherent (as shown in the dashed box). To provide a more intuitive example, (b) illustrates that the spatial domain detection task of a hypothetical cell, represented by a question mark, is established by integrating information from both gene expression (x_i) and neighbourhood configuration (c_{N_i}). The colour of each node indicates the expression levels of each cell, and the number in each node suggests the domain assignment. In this scenario, integrating these sources of information leads to the cell being assigned to the spatial domain "1" rather than "2" and "3" (Zhu et al., 2018).

BayesSpace

The spatial clustering BayesSpace (Zhao et al., 2021) model is a computational approach that leverages the neighbourhood structure present in spatially resolved transcriptomics data to enhance resolution at the subspot level. BayesSpace first employs a fully Bayesian statistics model using a Markov random field, and then it encourages spots of the same cluster to be near one another. This model takes inspiration from earlier spatial statistics techniques used in image analysis and microarray data (Besag, 1986; Gottardo et al., 2006). In contrast to earlier methodologies, BayesSpace offers greater flexibility in terms of specifying the clustering structure and error term.

The workflow of the BayesSpace technique is depicted in Figure 2.6. In this Figure labelled (a), after preprocessing the ST or Visium data, the data undergo spatial clustering based on similarities in their expression patterns. Through an enhanced clustering process, it is possible to refine these clusters further, resulting in a spatial map of higher resolution. Furthermore, this enhanced clustering technique enables higher-resolution predictions of gene expression, which can be utilized in the analysis of subsequent differential expression investigations. From (b) in this Figure, it is evident that ST and Visium spots are organized on square and hexagonal lattices, respectively, offering an intuitive method to establish a neighbourhood structure. We can define neighbours for each spot by shared edges. In addition, it is worth noting that every spot can be broken down into smaller subspots, which also have neighbours defined by shared boundaries.

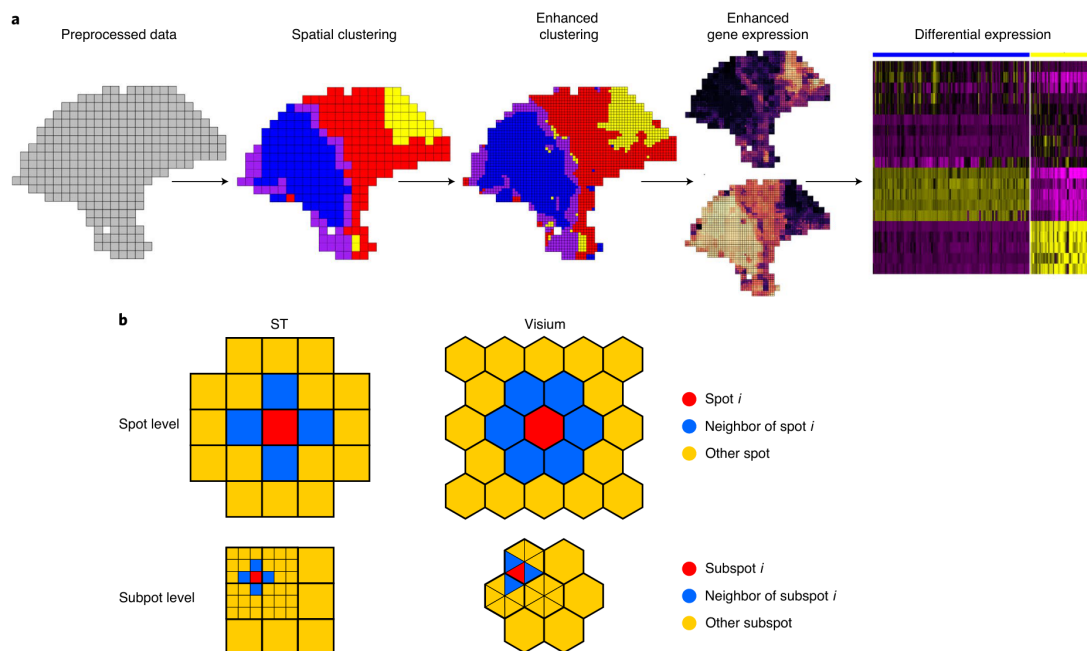


Figure 2.6: Schematic overview of BayesSpace

Source: Zhao, E., Stone, M. R., Ren, X., Guenthoer, J., Smythe, K. S., Pulliam, T., ... & Gottardo, R. (2021). Spatial transcriptomics at subspot resolution with BayesSpace. *Nature biotechnology*, 39(11), 1375-1384. <https://doi.org/10.1038/s41587-021-00935-2>

When we represent this model mathematically, we need to assume that for each spot i , a low d-dimension representation y_i , such as the principle components of the gene expression vector, can be derived. The model can then be described as:

$$(y_i|z_i = k, w_i) \sim N(y_i; \mu_k, w_i^{-1}\Lambda^{-1})$$

where $z_i \in \{1, \dots, q\}$ represents the latent cluster to which i is associated, μ_k refers to the average vector for spatial domain labels k , Λ means the precision matrix, and w_i stands for an unknown scaling factor specific to the observation. In this model, it is assumed that there exists an identical precision matrix among all clusters since the number of unknown parameters in the precision matrix grows rapidly with the increase in the number of clusters and principal components being modelled.

Parameters are estimated using the Markov chain Monte Carlo (MCMC) technique. By default, the value z can be initialized using a non-spatial clustering approach like the mclust method (Fraley et al., 2012). Subsequently, the values μ_k , Λ and w_i are updated using Gibbs sampling while the Metropolis-Hastings algorithm is employed to update each z_i in an iterative and sequential manner. It is worth noting that the update of each z_i considers both the likelihood and spatial prior data. The Markov random field prior, as provided by the Potts model, is presented below:

$$\pi(z_i) = \exp\left(\frac{\gamma}{\langle ij \rangle} \times 2 \sum_{\langle ij \rangle} I(z_i = z_j)\right)$$

where $\langle ij \rangle$ represents all spots j that are neighbours of i , I stands for the indicator function, and γ denotes a constant parameter that determines the smoothing strength. With this approach, neighbouring spots are more likely to be assigned in the same cluster (Zhao et al., 2021).

BANKSY

The Building Aggregates with a Neighbourhood Kernel and Spatial Yardstick (BANKSY) model is an approach that combines cell-type clustering with spatial domain segmentation to provide spatially informed clustering. Within the framework of the BANKSY algorithm, we focus on grouping a collection of cells, considering both the gene expressions and the spatial coordinates of the cells or spots. This is achieved by establishing a product space that consists of transcriptomes of cells and their corresponding neighbourhoods, which indicate the status of the cells and the microenvironment correspondingly (Singhal et al., 2022).

Figure 2.7 depicts how to incorporate the spatial microenvironment information into cell clustering and spatial domain detection. For a simple illustration using (a), we can consider an individual cell i and its neighbouring cells in physical space. The gene-cell matrix,

highlighted in purple, is augmented with a neighbourhood expression matrix shown in pink. The column associated with the i^{th} cell denotes the mean expression of its neighbouring cells. Furthermore, assuming there are only two single-cell types in the neighbour-enhanced space, namely Zones A and B, as depicted in (b), the BANKSY method utilizes spatial information (x and y) from the local microenvironment by calculating the average expression of neighbouring cells. This assists in distinguishing the two clusters, a task that might be challenging when relying only on a cell's own gene expression. Similarly, we can extend this idea to spatially resolved transcriptomics data, treating spots as individual “cells”. However, in practice, spots might consist of multiple cells. This approach enables us to use spatial information to identify distinct spatial zones or regions.

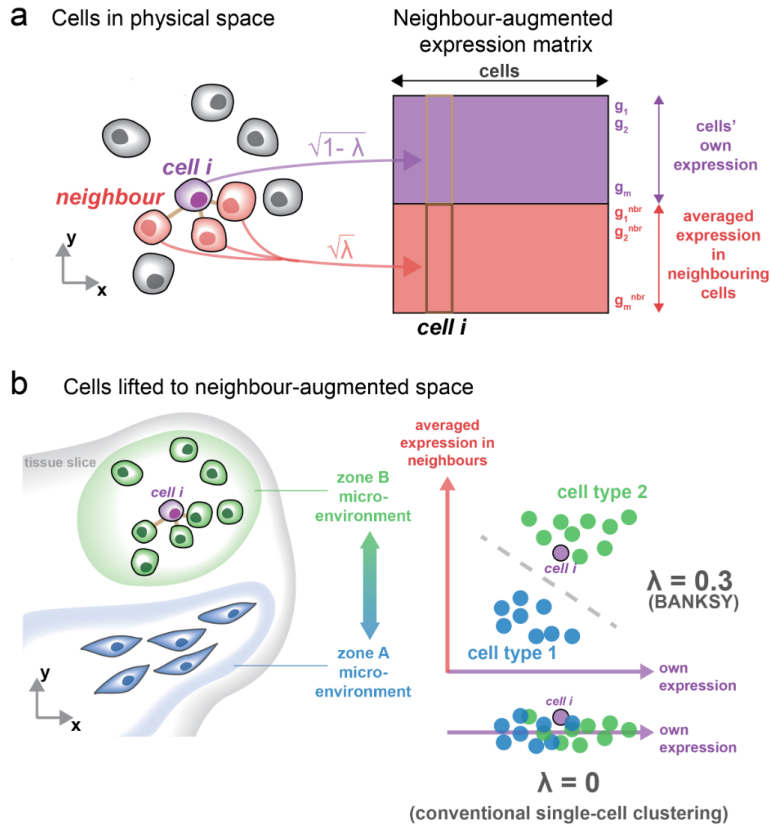


Figure 2.7: Schematic overview of BANKSY

Source: Singhal, V., Chou, N., Lee, J., Liu, J., Chock, W. K., Lin, L., ... & Prabhakar, S. (2022). BANKSY: a spatial omics algorithm that unifies cell type clustering and tissue domain segmentation. bioRxiv, 2022-04. <https://doi.org/10.1101/2022.04.14.488259>

If we mathematically represent this model, let us assume that a group of cells is spatially distributed. These cells are indexed by a set, $\mathcal{I} = \{1, 2, \dots, N\}$, where N represents the index for different cells. Each of these cells has a spatial coordinate, grouped together as $\chi = \{x_i \in \mathbb{R}^d | i \in \mathcal{I}\}$ within a d -dimensional space, commonly $d = 2$ or $d = 3$. Moreover, we assume that the expression of a specific set of genes indicated by $m \in \mathbb{Z}_{\geq 1}$ has been examined for each cell. This expression data can be represented as a gene-cell expression matrix, denoted as $H = [h_1 \ h_2 \ \dots \ h_N] \in \mathbb{R}^{m \times N}$. Here, $h_i \in \mathbb{R}^m$ represents the expression

levels of the m genes in cell i . Then, neighbouring cells have been allocated to every cell based on their relative spatial locations. In other words, a neighbourhood is related to the i -th cell.

Furthermore, let us represent $w_{ij} \in R$ as the weight related to the edge from cell j to index cell i , and let $r_{ij} \in R$ denote the physical distance between these cells (measured in any units, given that w_{ij} is dimensionless). The following equation is about w_{ij} ,

$$w_{ij} = \frac{1/r_{ij}}{\sum_{q=1}^k 1/r_{iq}}$$

where the denominator guarantees that the weights corresponding to the neighbourhood of a cell sum to 1. The normalization process with other weighting kernels follows a similar pattern. Based on this framework, we can calculate the average signature of the neighbourhood linked to the i -th cell as

$$\gamma_i = \sum_{j \in N_i} w_{ij} c_j$$

Then, we combine these neighbourhood expression vectors to form a neighbour expression gene-cell matrix,

$$\Gamma = [\gamma_1 \ \gamma_2 \ \dots \ \gamma_N] \in R^{m \times N}$$

and merge C and Γ to form the neighbour-augmented matrix, often referred to as the BANKSY matrix,

$$B = \begin{bmatrix} \sqrt{1-\lambda} \cdot C \\ \sqrt{\lambda} \cdot \Gamma \end{bmatrix} \in R^{2m \times N}$$

where λ ranges between 0 and 1, determining the relative importance of the two-component matrices.

Graph-based clustering approaches, like the Louvain (De Meo et al., 2011) or Leiden (Traag et al., 2019) methods, have been increasingly popular in genomics research given the presence of a large number of cells. These algorithms are particularly effective when applied to feature-object matrices, as illustrated in the above BANKSY matrix B . Consequently, the BANKSY method selects the Leiden algorithm by default to implement the spatial domain detection task (Singhal et al., 2022).

2.3.1.2 2D deconvolution-based spatial clustering methods

For 2D spatial domain detection, beyond simply considering the spatial coordinates of the X-axis and Y-axis for clustering, some methods also incorporate cell type information. These approaches aim to decompose the cell-type composition information across locations. Afterwards, we can apply classical clustering methods, such as the K-means algorithm, to determine the spatial domains. We refer to these techniques as deconvolution-based spa-

tial clustering methods. There are various deconvolution methods, such as conditional autoregressive-based deconvolution (CARD) (Ma and Zhou, 2022), robust cell type decomposition (RCTD) (Cable et al., 2022), cell2location (Kleshchevnikov et al., 2022), and spatial dampened weighted least squares (spatialDWLS) (Dong and Yuan, 2021). In this thesis, we mainly focus on utilizing CARD and RCTD deconvolution techniques to identify spatial domains.

CARD

The Conditional Autoregressive-based Deconvolution (CARD) (Ma and Zhou, 2022) is designed to decompose cell types for spatial transcriptomics with spatial information taken into account. This method integrates cell-type-specific expression data from scRNA-seq with the correlation discovered in cell-type composition across different tissue locations. The incorporation of spatial correlation modelling enables the utilization of cell-type composition across different locations, hence enhancing the accuracy of deconvolution even in cases where there is a discrepancy with scRNA-seq reference.

Figure 2.8 demonstrates the schematic of the CARD model. Using scRNA-seq data with cell-type-specific gene expression details (on the left) and spatial transcriptomics data featuring location information (on the right) as input, CARD employs a non-negative matrix factorization approach to deconvolve and generate estimated proportions of cell types in different spatial locations.

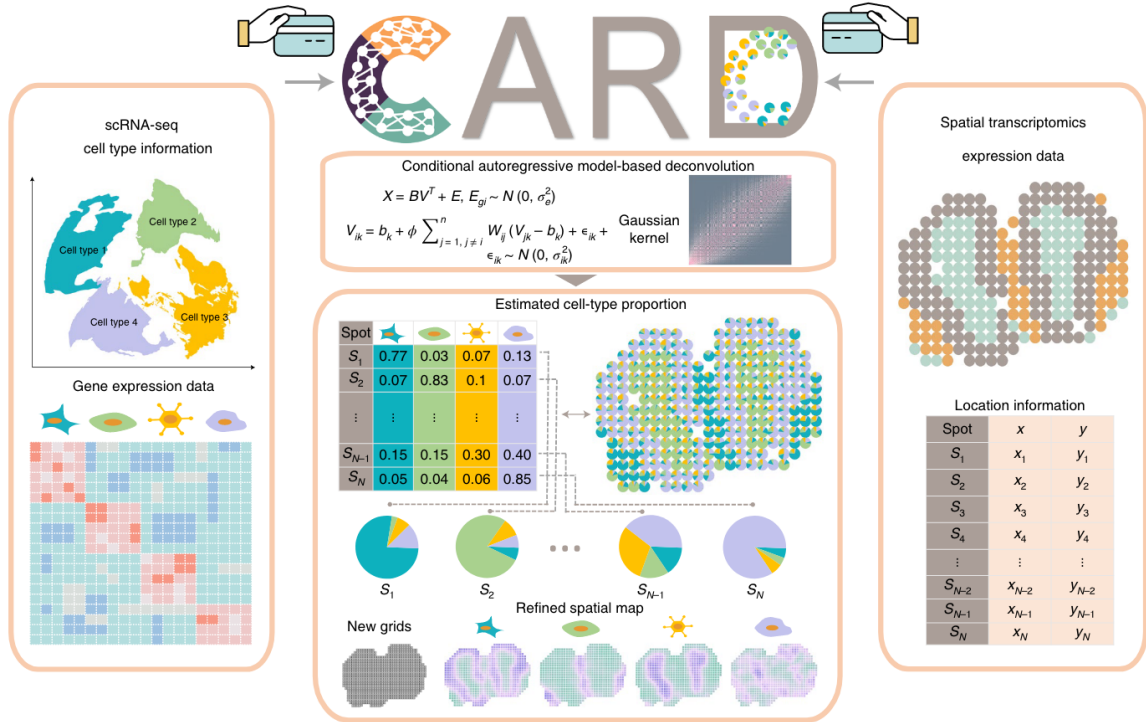


Figure 2.8: Schematic overview of CARD

Source: Ma, Y., & Zhou, X. (2022). Spatially informed cell-type deconvolution for spatial transcriptomics. *Nature biotechnology*, 40(9), 1349-1359. <https://doi.org/10.1038/s41587-022-01273-7>

A distinct characteristic of CARD is its capability to integrate the spatial correlation patterns in cell-type composition throughout tissue locations by the assumption of conditional autoregressive (CAR) modelling (Banerjee et al., 2014; Lee, 2011), with the mathematical formula shown below:

Firstly, a non-negative matrix factorization model can be expressed as follows:

$$X = BV^T + E$$

where each component in the V matrix is required to have non-negative values, and E is a J -by- T matrix of residual errors, where each component independently and identically follows a normal distribution $E_{gi} \sim N(0, \sigma_e^2)$.

Then, we apply a CAR modelling assumption on each column of the matrix V . Specifically, for the g^{th} column corresponding to a cell type, we can assume the following equation:

$$V_{ig} = b_g + \phi \sum_{j=1, j \neq i}^n W_{ij}(V_{ig} - b_g) + \epsilon_{ig}$$

where V_{ig} denotes the proportion of cell type g at the i^{th} location. The g^{th} cell-type-specific intercept, denoted as b_g , gives an average representation of cell-type composition across various locations, the weight matrix W , which is an $T \times T$ non-negative matrix, contains elements W_{ij} that provide the weightage for deriving the cell-type composition at the i^{th} location using the composition data from the j^{th} location. The parameter ϕ serves as an index for spatial autocorrelation, determining the strength of the spatial correlation for cell-type compositions. Lastly, the residual error, ϵ_{ig} , adheres to a normal distribution, represented as $\epsilon_{ig} \sim N(0, \sigma_{ig}^2)$. The CAR model assumption regarding the composition of the g^{th} cell type, V_{ig} , on the i^{th} location can be expressed as a weighted sum of the g^{th} cell type compositions on all other locations, $V_{jg}(j \neq i)$. As a result, the assumption made by the CARD model regarding the matrix V enables us to utilize information from several locations to estimate the composition of cell types in specific locations.

Hence, a notable advantage of CARD is its ability to harness the spatial correlation structure, ensuring precise and robust deconvolution of spatially resolved transcriptomics data. Furthermore, by integrating spatial correlation, CARD can infer both cell-type compositions and gene expression levels in new tissue locations, which enables the creation of a refined spatial map with adaptable high resolution, suitable for various spatial transcriptomics technologies (Ma and Zhou, 2022).

RCTD

The robust cell type decomposition (RCTD) (Cable et al., 2022) method is another well-known deconvolution method, which utilizes cell type profiles acquired from single-cell RNA-seq to effectively decompose mixtures of cell types while also correcting variations in-

roduced by different sequencing technologies. Specifically, RCTD is a supervised learning model that aims to break down mixtures of RNA sequencing data into single cell types, which facilitates the identification and allocation of various cell types to spatial transcriptomic spots. This method applies a statistical model to estimate combinations of cell types at every spot.

As shown in Figure 2.9, the RCTD model's entire process is depicted. The model takes as input a scRNA-seq dataset labelled by cell type and a spatial transcriptomics dataset with unidentified cell types. Then, RCTD employs a probabilistic model based on the scRNA-seq reference to deduce cell types in individual spots, which might contain mixed cell types. Following that, RCTD predicts the maximum-likelihood cell-type proportions. In doublet mode, RCTD restricts each spot to a maximum of two cell types. As an alternative, RCTD can also use all cell types to determine the optimal fit for a spot. The final output is a spatial map of cell types, where transparency levels indicate the estimated cell type proportions.

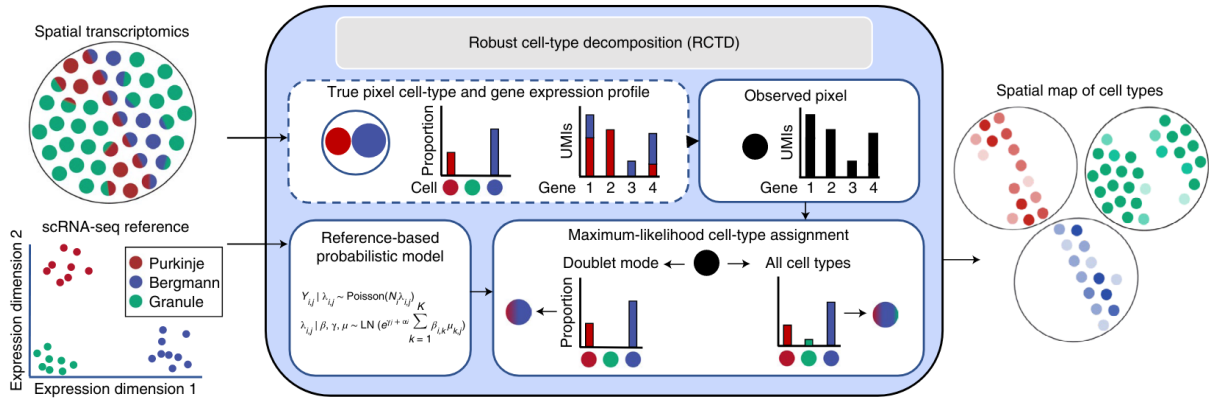


Figure 2.9: Schematic overview of RCTD

Source: Cable, D. M., Murray, E., Zou, L. S., Goeva, A., Macosko, E. Z., Chen, F., & Irizarry, R. A. (2022). Robust decomposition of cell type mixtures in spatial transcriptomics. *Nature biotechnology*, 40(4), 517-526.

<https://doi.org/10.1038/s41587-021-00830-w>

Next, let us investigate the statistical model of RCTD to identify a mixture of cell types. Consider each spot $i = 1, \dots, I$ in the spatially resolved transcriptomics data. Here, $Y_{i,j}$ represents the observed gene expression counts for each gene $j = 1, \dots, J$. The subsequent hierarchical model can be employed to model these counts:

$$Y_{i,j} | \lambda_{i,j} \sim \text{Poisson}(N_i \lambda_{i,j})$$

$$\log(\lambda_{i,j}) = \alpha_i + \log\left(\sum_{g=1}^G \beta_{i,g} \mu_{g,j}\right) + \gamma_j + \epsilon_{i,j}$$

where N_i denotes the total transcript count or the number of UMIs associated with spot i . G stands for the total cell types identified in our dataset. The parameter α_i means a

predetermined effect specific to spot i . Moreover, $\mu_{g,j}$ denotes the average gene expression profile for the cell type g , and gene j , $\beta_{i,g}$ reflects the proportion indicating the contribution of cell type g to spot i . The γ_j term is related to a random effect specific to the gene platform, and $\epsilon_{i,j}$ is a random effect to adjust for additional variation sources, including spatial influences.

The incorporation of Poisson noise modelling enables RCTD to address the issue of sampling noise effectively. The primary aim of this mode is to estimate the $\beta_{i,g}$ values, which indicate the presence of one or more cell types in each spot i . These values are subject to the constraint that $\sum_{g=1}^G \beta_{i,g} = 1$, and each $\beta_{i,g}$ value is greater than or equal to 0 (Cable et al., 2022).

2.3.2 3D spatial clustering methods

The methods mentioned above are restricted to a single tissue slice in the 2D plane. However, many spatial transcriptomic studies usually gather multiple neighbouring sections from the same tissues or obtain tissue samples from various individuals. Sections obtained from identical or analogous tissues often exhibit a similar spatial domain and cell type composition. By jointly modelling multiple tissue sections or samples together, we can leverage information from one sample to another, potentially improving the performance of cell type clustering and spatial domain identification in spatially resolved transcriptomics data (Li and Zhou, 2022). Therefore, we can go beyond 2D and reconstruct a comprehensive 3D tissue structure, which is more biologically meaningful. In this section, we will mainly discuss two 3D methods, one is Bayesian Analytics for Spatial Segmentation (BASS) (Li and Zhou, 2022) and the other is STitch3D (Wang et al., 2023a).

BASS

The Bayesian Analytics for Spatial Segmentation (BASS) (Li and Zhou, 2022) employs a Bayesian hierarchical modelling technique to conduct integrated transcriptomic analyses at multiple scales. This approach encompasses joint clustering of cell types and detecting spatial domains, allowing both analytical tasks to be performed jointly. In both these analyses, BASS effectively incorporates the spatial correlation structure and seamlessly combines gene expression data with spatial location data to enhance their accuracy and effectiveness. Furthermore, BASS possesses the ability for multi-sample analysis, which involves jointly modelling multiple tissue slices, enabling the construction of a 3D representation of tissue structure in spatially resolved transcriptomics data.

Figure 2.10 illustrates the overview of the BASS model. The BASS algorithm takes as input the gene expression matrix and spatial location data from multiple tissue sections, integrating both pieces of information within a hierarchical Bayesian framework. During the analysis process, BASS regards the cell type label (g) and the spatial domain label

(k) for each cell inside the tissue as latent variables. These variables are deduced using an efficient inference algorithm.

Specifically, by incorporating both expression and location information, we employ the below three equations to establish a hierarchical modelling framework that captures the relationship among gene expression features, cell type labels, spatial domain labels, cell type compositions, and neighbourhood graphs:

$$x_i^{(l)} | g_i^{(l)} = g \sim \text{Normal}(\mu_g, \Sigma)$$

$$g_i^{(l)} | k_i^{(l)} = r \sim \text{Cat}(\pi_r)$$

$$k^{(l)} \sim \text{Potts}(V^{(l)}, \beta)$$

Following the analysis, BASS generates outputs that consist of the assigned labels for cell type clusters, spatial domain labels, and the proportions of each cell type within each spatial domain. Based on this output information, we can accurately obtain the results of spatial domain detection after 3D reconstruction, which helps us identify different tissue regions to discover further biological insights (Li and Zhou, 2022).

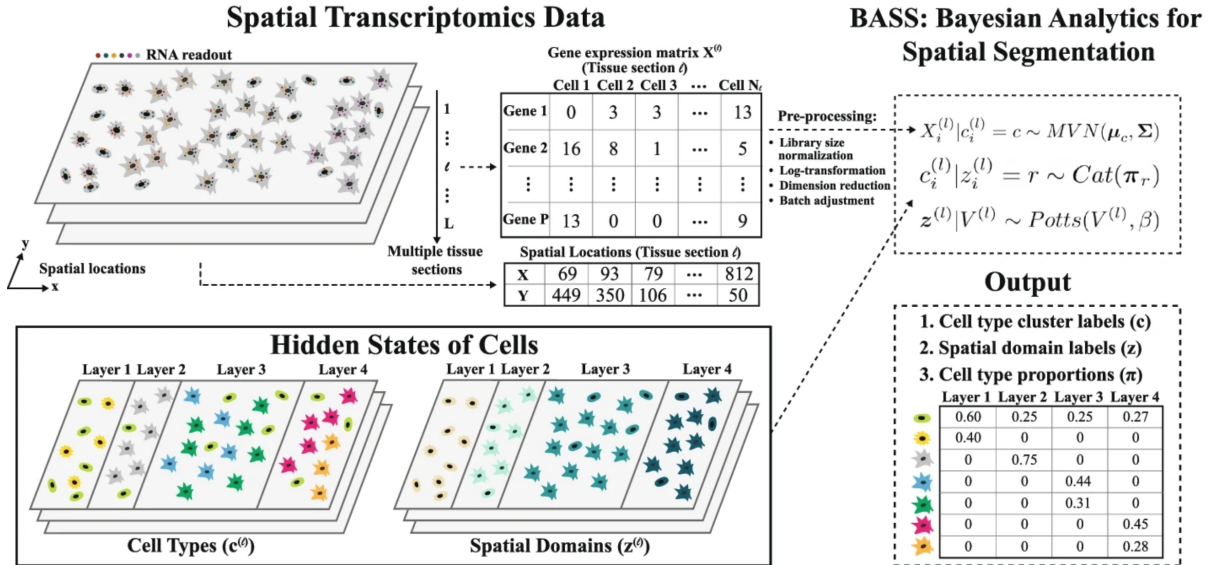


Figure 2.10: Schematic overview of BASS

Source: Li, Z., & Zhou, X. (2022). BASS: multi-scale and multi-sample analysis enables accurate cell type clustering and spatial domain detection in spatial transcriptomic studies. *Genome biology*, 23(1), 168.
<https://doi-org.ezproxy.library.sydney.edu.au/10.1186/s13059-022-02734-7>

STitch3D

The STitch3D (Wang et al., 2023a) technique is another advanced 3D method that aims to analyze 3D tissue structures using multiple tissue slices, focusing on two primary tasks: identifying spatial domains in 3D and decomposing the proportions of cell types in a 3D context. Through the simultaneous analysis of gene expression levels and spatial

coordinates across multiple adjacent tissue slices, STitch3D can differentiate biological variations within slices from batch-related effects. This technique enables the efficient integration of information across slices, facilitating the construction of biological meaningful 3D tissue models. Figure 2.11 displays the schematic workflow of STitch3D.

STitch3D requires specific input data, which includes gene expression matrices that are spatially resolved, containing 2D spatial coordinates extracted from several ST slices. Additionally, these matrices should be incorporated with cell-type-specific gene expression profiles derived from a paired scRNA-seq reference dataset. Then, the preprocessing of STitch3D involves two essential procedures: first, aligning spots from different tissue slices to establish 3D coordinates for each spot, and second, constructing a comprehensive 3D global graph.

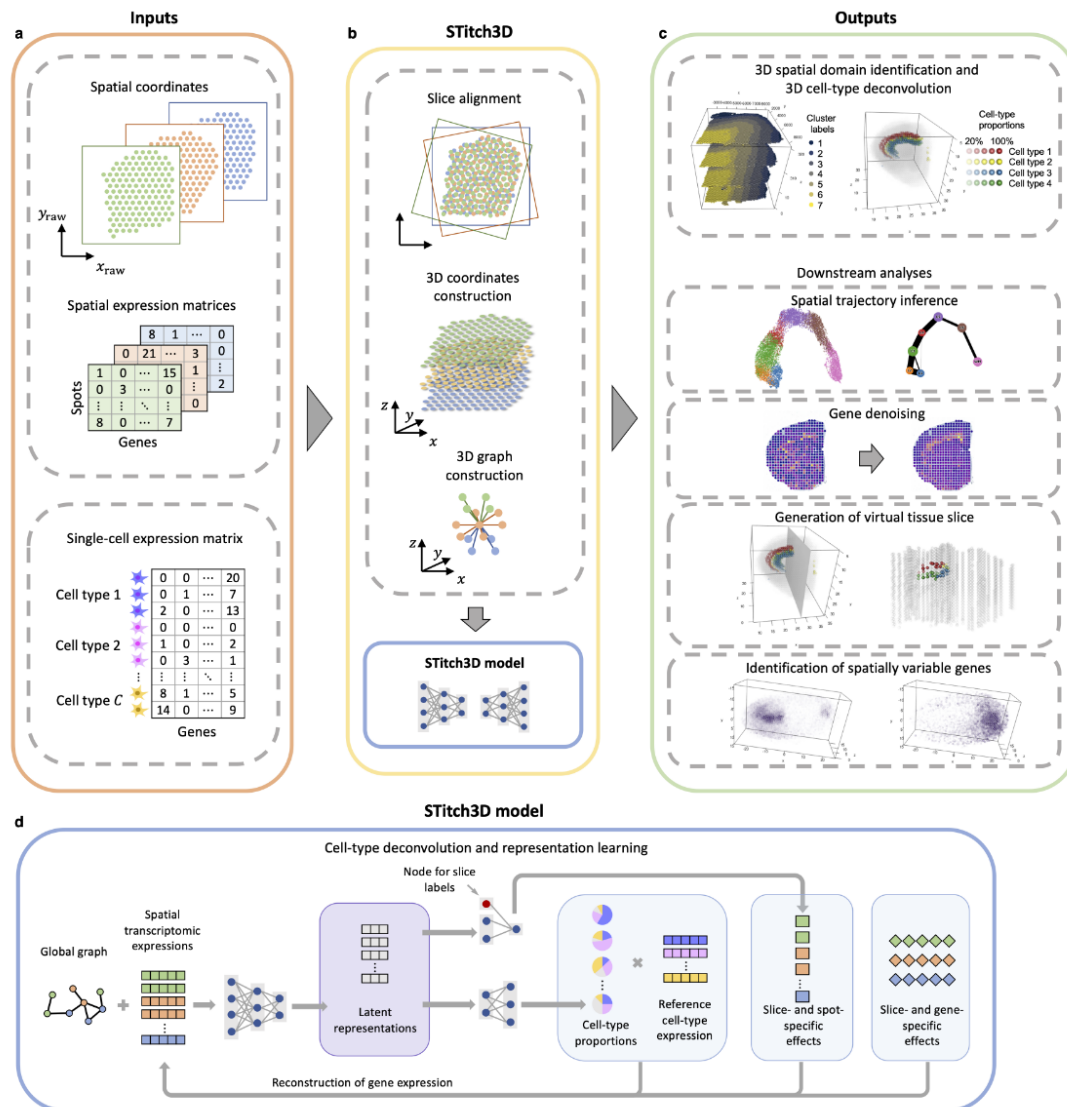


Figure 2.11: Schematic overview of STitch3D

Source: Wang, G., Zhao, J., Yan, Y., Wang, Y., Wu, A. R., & Yang, C. (2023). Construction of a 3D whole organism spatial atlas by joint modeling of multiple slices. bioRxiv, 2023-02. <https://doi.org/10.1101/2023.02.02.526814>

Subsequently, the STitch3D method employs a graph attention-based neural network to capture latent representations of both spatial spots and cell-type proportions while integrating crucial 3D spatial information. The output generated by STitch3D includes 3D spatial domain detection results and the estimated proportions of distinct cell types within tissue structures (Wang et al., 2023a). This information can be utilized for various downstream analyses, such as spatial trajectory inference, etc.

2.3.3 Challenges in spatial clustering analysis

The challenges of spatial clustering analysis are similar to those of scRNA-seq clustering analysis. For example, both have the problem of the “curse of dimensionality”. Although comprehensive benchmarking studies have been conducted to evaluate different methods for scRNA-seq clustering methods, the field of spatial clustering has not received the same level of scrutiny. Despite the plethora of spatial clustering approaches available, there is a noticeable lack of comparative analysis to investigate the superior performance of one approach over another. The existence of this gap is a significant challenge in the advancement of spatial clustering research.

2.4 Summary

In this chapter, we conducted an in-depth review of the literature on scRNA-seq clustering analysis, elucidating classical methods and related challenges. We then reviewed spatial clustering techniques designed for spatially resolved transcriptomics data in 2D and 3D contexts and their associated challenges. In subsequent experiments, we will apply these clustering algorithms to evaluate the performance of different clustering methods.

Chapter 3

Two-dimensional Spatial Domain Detection

This chapter delves into two-dimensional (2D) spatial domain detection in spatially resolved transcriptomics data. It begins with a detailed presentation of the real dataset in Section 3.1, followed by a comprehensive description of the experimental design in Section 3.2. The subsequent portion involves conducting simulation studies. In this context, the chapter introduces two simulation methodologies, namely CARD and scDesign3. Through these simulation methods, the analysis explores the underlying patterns, with a particular focus on circular and regional patterns, extending to the analysis of real human heart data. Following this, the chapter provides a detailed overview of the evaluation procedure, leading to a discussion of the core results in Section 3.3. Finally, the chapter concludes with a summary that encapsulates the key findings and discussions presented in Section 3.4.

3.1 Data description

Human heart dataset A recent study produced a spatiotemporal depiction of the gene expression pattern during the development of the human heart. To be more specific, multiple slices of human embryonic heart tissue samples gathered at 4.5–5, 6.5, and 9 post-conception weeks (PCW) were stained using the immunohistochemistry (IHC) method to reveal the spatiotemporal perspective of human heart development ([Asp et al., 2019](#)). For each stage of data, multiple slices can be generated utilizing spatial transcriptomics (ST) technology. This chapter will mainly focus on the dataset of the human heart at 6.5 PCW.

3.2 Experimental design

Figure 3.1 illustrates the experimental design section of this chapter. Initially, we have data including predefined patterns, specifically the circle and region patterns and patterns from a real dataset. These data are then input into the simulators, CARD and scDesign3. Subsequently, we conduct a model comparison containing three primary categories of clustering methods, divided as follows:

1. **Classical clustering:** Using the gene expression data, we employ algorithms such

as K-means, Hierarchical clustering, and Louvain community detection.

2. **2D spatial clustering:** By integrating spatial expression with x and y coordinates, spatial clustering methods such as HMRF, BayesSpace, and BANKSY are utilized.
3. **2D deconvolution-based spatial clustering:** This category synthesizes three types of data: spatial expression, x and y coordinates, and cell type information, with the methods CARD and RCTD being applied.

Finally, we evaluate the performance of each clustering algorithm using the metrics of the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) compared to predetermined patterns, which are assumed to be the ground truth.

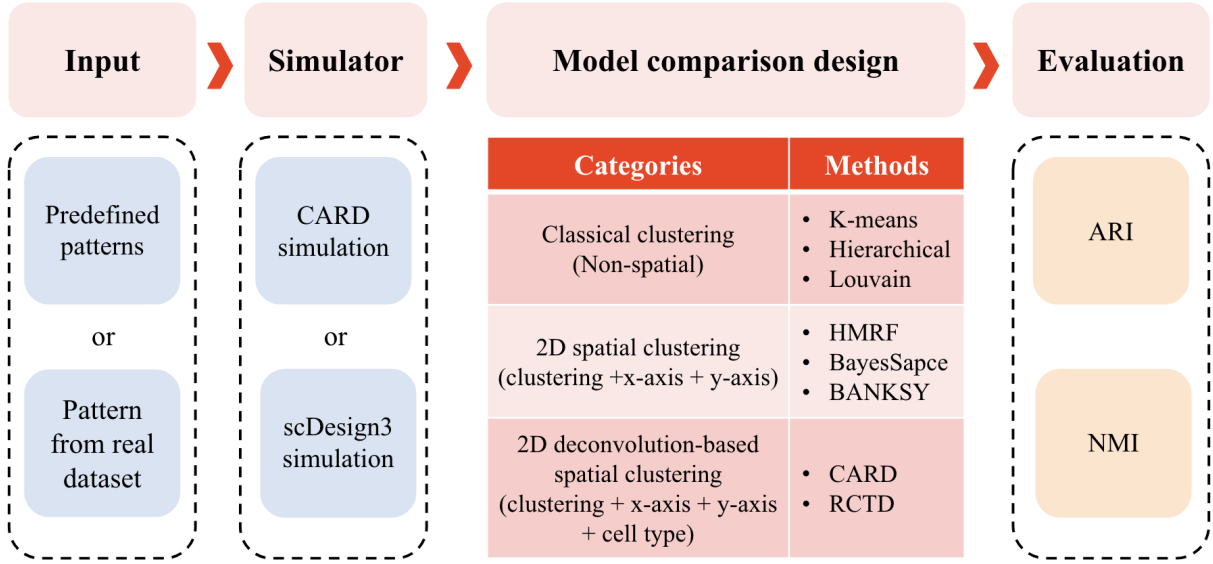


Figure 3.1: Overview of experimental design

3.3 Simulation studies

3.3.1 Simulation methods: CARD and scDesign3

For the simulator, we have two methods. The first approach, as described in the CARD (Ma and Zhou, 2022) paper, produces spots that are not spatially related. The second method is from scDesign3 (Song et al., 2023), which generates spatially correlated spots. In the following descriptions of simulation methods, we will use the notation defined in Chapter 2 for representation.

CARD simulation

Figure 3.2 presents a detailed depiction of the steps involved in CARD simulation.

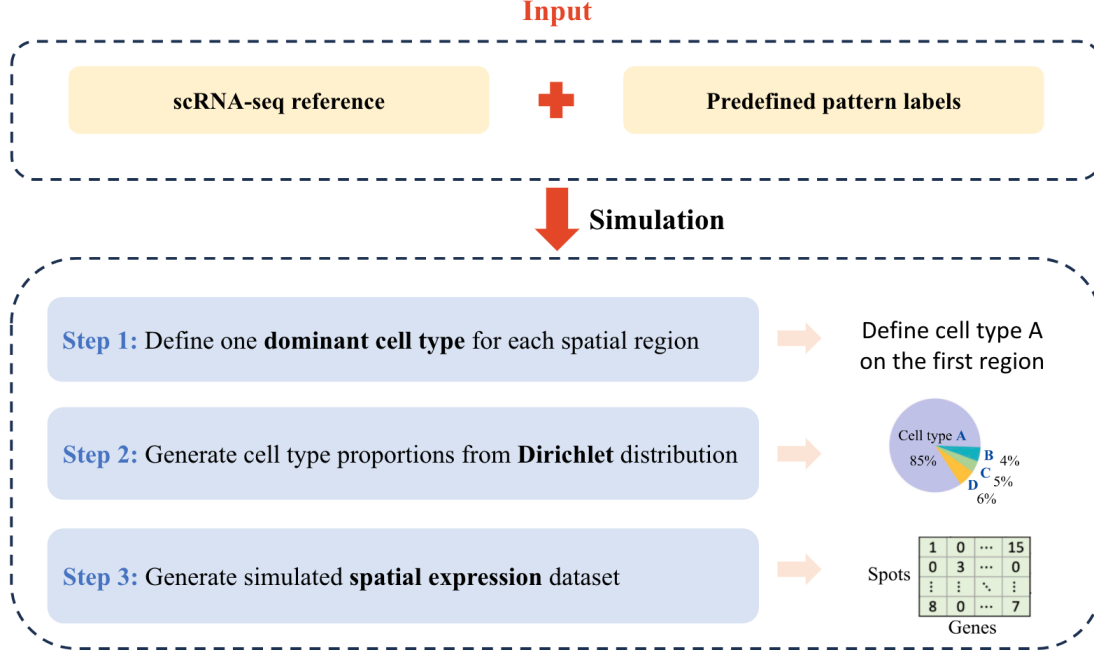


Figure 3.2: CARD simulation workflow

The two primary inputs are a scRNA-seq reference and predefined pattern labels in spatial transcriptomics:

1. **scRNA-seq reference:** Matrix M_{sc} of dimensions $p \times q$, where p denotes the number of genes and q denotes the number of cells, and g distinct cell types.
2. **Predefined pattern labels:** In different predefined patterns, k spatial regions are identified.

Subsequently, inputs are put into the CARD simulation, involving three specific steps:

- **Step 1:** Define one dominant dominant cell type for each spatial region. For example, we can define cell type A as the first spatial region.
- **Step 2:** Generating cell type proportions for each spot by utilizing the Dirichlet distribution. Although the proportions of cell types may differ across different spots, the dominant cell type remains as previously defined. As shown in Figure 3.2, the dominant cell type proportion is still cell type A.
- **Step 3:** From the determined cell type proportions for each spot, the simulation procedure samples from a reference scRNA-seq dataset to create a synthetic spatial dataset represented as a spot-by-gene matrix.

In summary, the process determines cell type proportions for each spot and then creates a synthetic spatial dataset based on these proportions. However, the lack of spatial correlation in each spot due to the independence of the simulation process may lead to potential inaccuracies in capturing the true characteristics of the real dataset.

scDesign3 simulation

The second simulation method is the scDesign3 simulation method (Song et al., 2023), which can generate spot-resolution spatial transcriptomics data by incorporating spatial correlation among different spots. This method is more sophisticated than the CARD simulation method since it considers spatially correlated information for each spot throughout the simulation procedure. Figure 3.3 depicts the essential processes performed in the scDesign3 simulation.

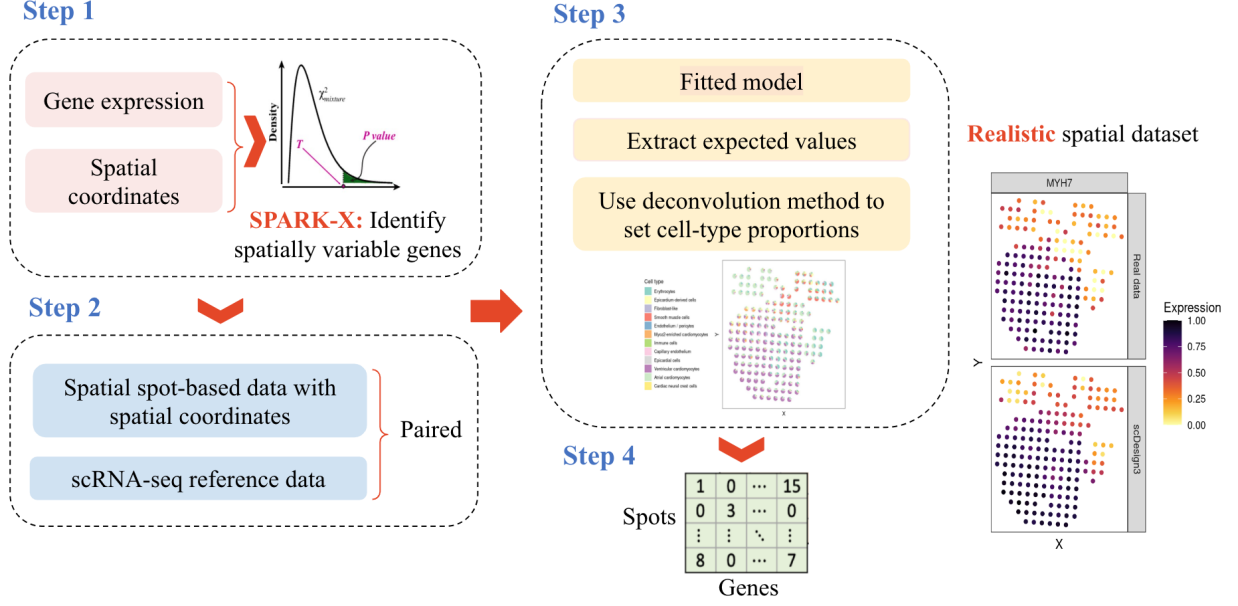


Figure 3.3: scDesign3 simulation workflow

- Step 1: Selecting SVGs** Due to the computational limitations of the scDesign3 simulation, we select spatially variable genes using the spatially variable gene (SVG) detection technique before conducting the simulation. In general, selecting 1000-1500 SVGs provides sufficient information for subsequent spatial domain detection tasks (Chen et al., 2022). To optimize computational efficiency, we employ the SPARK-X SVG technique to select the top 1000 spatially variable genes, as expressed as M'_{sp}
- Step 2: Paired dataset** Following identifying the top 1000 SVGs, we utilize this selected spatial gene expression matrix M'_{sp} and paired it with the scRNA-seq reference to initiate the scDesign3 simulation process.
- Step 3: Implementing simulation process** Firstly, we employ scDesign3 to estimate the cell-type reference from scRNA-seq data. Following this, we obtain the fitted models for scRNA-seq and spatial data and then extracted their expected values. Subsequently, the CIBERSORT deconvolution method (Newman et al., 2015)

is utilized to decompose each spot’s expected expression into cell-type proportions. The purpose of this stage is to establish the true proportions of cell types.

- **Step 4: Generating simulated spatial expression dataset** Afterward, we simulate new spatial data by creating spots, where each spot represents the average of 50 cells divided by 5, yielding an average of 10 cells per spot. The spatial data will become smoother by growing the number of cells, which conforms more closely with the anticipated spatial expression. Subsequently, the spatial gene expression matrix was produced, with each row representing a spot and each column denoting the corresponding gene expression for that spot.

Finally, we can examine the simulated results. On the right side of Figure 3.12, we take the top SVG “MYH7” as an example and observe that the expression levels of real and scDesign3 simulated data are nearly identical, indicating that the scDesign3 simulation effectively captures the characteristics of the real dataset.

3.3.2 Input patterns

Circle pattern

Firstly, we generate a three-layer concentric circle pattern at a fixed centroid using specified radii and the number of points per layer. By using the CARD simulation method described above, we obtained the following results from one simulation, as displayed in Figure 3.4. From the results, the dominant cell type in each layer aligns with our predefined cell type: the first layer is “Smooth muscle cells”, the second layer is “Immune cells”, and the third layer is “Epicardial cells”. For every spot, we have six different cell types, but their cell type proportions are different. Following this, we can obtain a spatial expression dataset by utilizing the cell-type proportions.

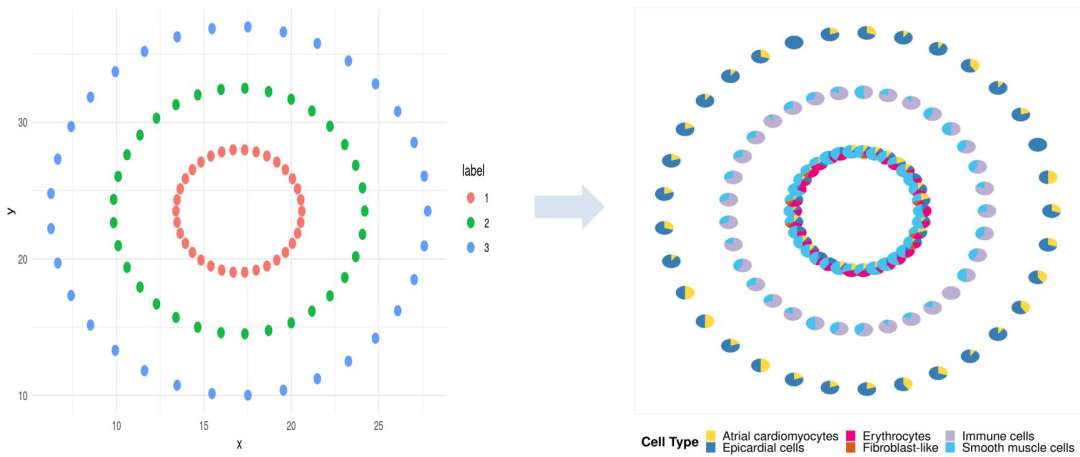


Figure 3.4: Circle pattern simulation results

Region pattern

Following a similar approach to pattern one, we initially obtained pattern two’s result from the CARD simulation, as depicted in Figure 3.5. The simulation results show that each region’s primary cell type matches our predefined dominant cell types: “Smooth muscle cells” predominate in the first region, “Immune cells” predominate in the second, and “Epicardial cells” predominate in the third. Six different cell types are present in each spot, although their proportions differ. After that, we can use the cell-type proportions to generate a spatial expression dataset.

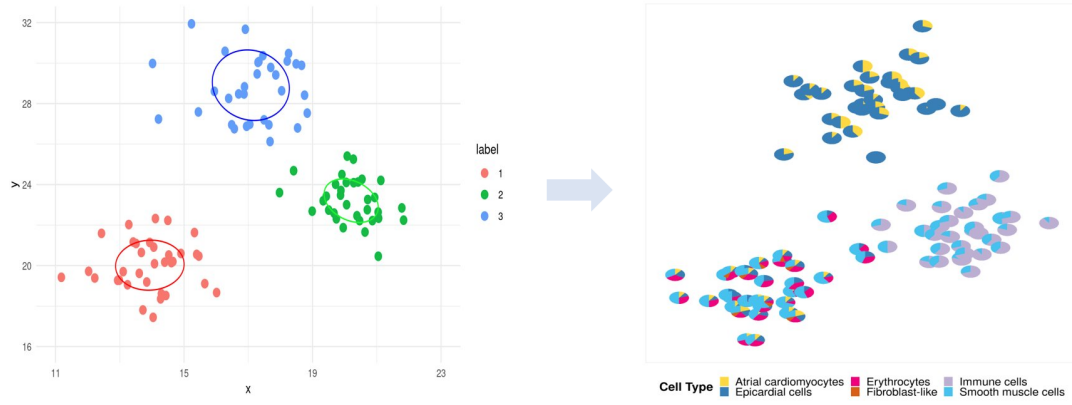


Figure 3.5: Region pattern simulation results

Real data pattern

Following the analysis of the simulated dataset, we move forward with implementing (spatial) clustering methods using the real human heart dataset. Our challenge is to obtain the real data pattern, also referred to as ground truth. The essential steps to determine the ground truth are depicted in Figure 3.6.

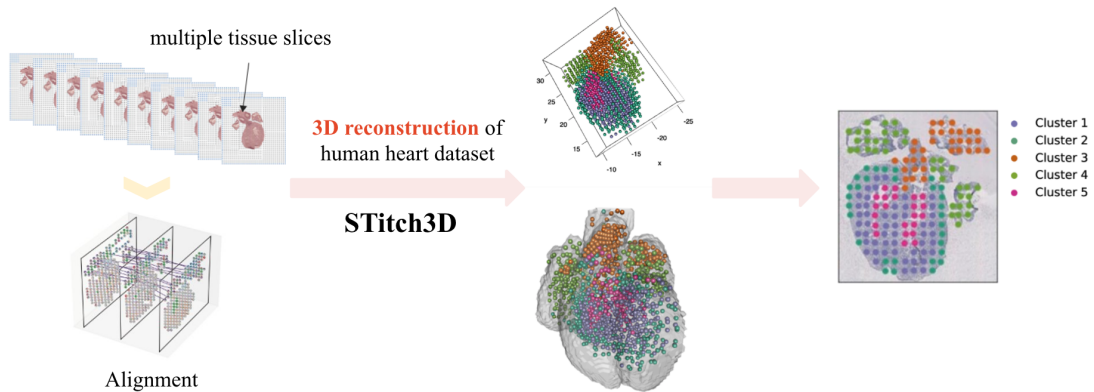


Figure 3.6: Predefined pattern for real data: human heart dataset

The 6.5 PCW human heart dataset comprises nine tissue slices from slice 0 to slice 8. We can apply the primary alignment method discussed in Chapter 2 to align the multiple

slices. Then, we employed the STitch3D technique to reconstruct the anatomical structure of the human heart in three dimensions. Our goal was to utilize this 3D model to accurately capture five distinct spatial regions resembling the actual biological tissue structure. For instance, cluster 3 identified the outflow tract (OFT), while the left and right atrium were accurately classified under cluster 4 (Wang et al., 2023a). To conduct a more comprehensive analysis, we selected slice 6 as the focal point to evaluate and compare the performance of different clustering methods.

Following this, the CARD or scDesign3 simulation methods can be utilized to determine different proportions in each spot and then generate a simulated spatial expression dataset, as depicted in Figure 3.7.

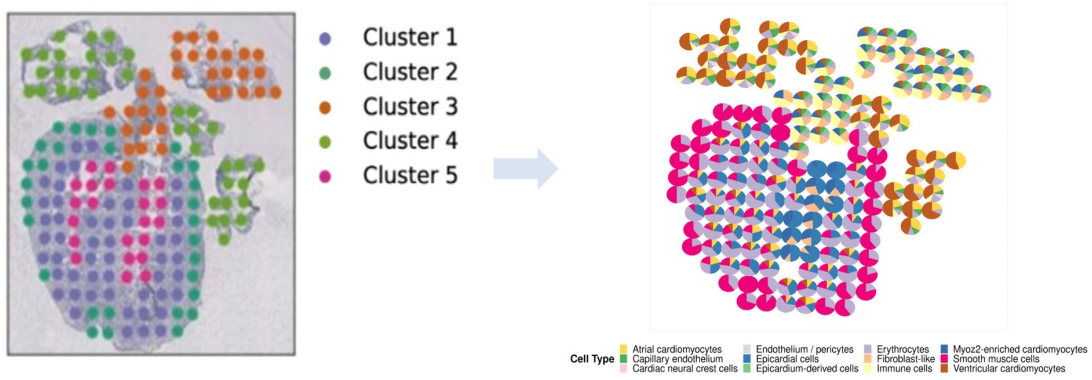


Figure 3.7: Real data simulation results

3.3.3 Evaluation procedure

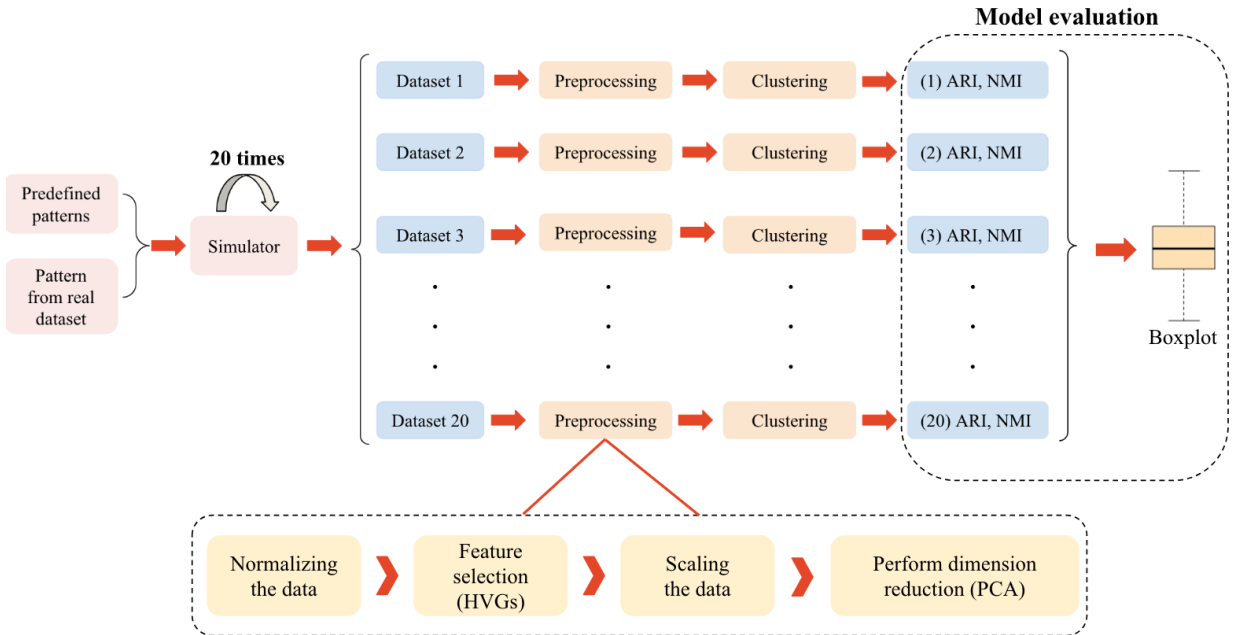


Figure 3.8: Evaluation procedure

Figure 3.8 illustrates the workflow of our evaluation procedure. After obtaining the input pattern, we can utilise simulators like CARD or scDesign3 to conduct experiments 20 times, yielding 20 experimental datasets, and then proceed with analysis following the prescribed steps at the bottom of the Figure. The preprocessing steps are implemented using the standard Seurat workflow, which consists of the following four steps:

- **Step 1 Normalising the data:** Using the global-scaling normalisation method “LogNormalize” by centring and scaling the simulated data matrix.
- **Step 2 Feature selection (HVGs):** The feature selection is conducted to identify highly variable genes (HVGs). Concentrating on a group of genes that show significant variation between cells in the dataset can enhance the detection of biological signals in subsequent single-cell analysis (Brennecke et al., 2013). We select 2,000 genes for each simulated data set, and these 2,000 genes will be used in the upcoming PCA analysis.
- **Step 3 Scaling the data:** After detecting HVGs, a linear “scaling” transformation is a required preprocessing step before using the PCA method to transform a vast sparse matrix into a dense format efficiently (DeBruine et al., 2021). To ensure that the average gene expression among cells is 0, we first shift the gene expression. Additionally, we scale gene expression to make cell-to-cell variation equal to 1.
- **Step 4 Perform dimension reduction (PCA):** We perform the PCA technique on the scaled simulated spatial expression data. In each simulation, 20 principle components(PCs) are chosen to ensure that PCs capture the most essential information from spatial gene expression.

After data preprocessing, we conduct spatial domain detection for each simulated dataset using three different clustering algorithms:

1. **Classical clustering:** For classical clustering approaches like K-means, Hierarchical, and Louvain methods, we incorporate only the 20 PCs into the models.
2. **2D spatial clustering:** For 2D spatial clustering techniques like HMRF, BayesSpace, and BANKSY, we incorporated the 20 PCs and the spatial coordinates from the x and y axes.
3. **2D deconvolution-based spatial clustering:** We initially use CARD and RCTD techniques to determine the cell type proportion for each spot. Using this information, we perform a K-means clustering analysis to identify the spatial domain labels for each spot.

To evaluate the performance of different clustering algorithms across the three distinct patterns, we utilised two evaluation matrices: adjusted Rand index (ARI) and normalised

mutual information (NMI). These metrics serve to quantitatively measure the concordance between clustering results on each simulated dataset and their respective predefined patterns.

Let S be a set of N cells. A clustering, denoted as $\mathbf{U}$, partitions S into non-overlap subset $\{\mathcal{U}_1, \mathcal{U}_2, \dots, \mathcal{U}_R\}$, where $\cup_{i=1}^R \mathcal{U}_i = S$ and $\mathcal{U}_i \cap \mathcal{U}_j = \emptyset$ for $i \neq j$. Then, we define $\mathbf{U} = \{\mathcal{U}_1, \mathcal{U}_2, \dots, \mathcal{U}_R\}$ as the gold standard cell type labels, $\mathbf{V} = \{V_1, V_2, \dots, V_R\}$ represents a partition obtained through a specific clustering method.

Using pair counting metrics, we can evaluate the item pairs where partitions $\mathbf{U}$ and $\mathbf{V}$ either agree or disagree. In detail, among the $\binom{N}{2}$ possible pairs in S , they can be categorized into one of these four types:

- (i) N_{11} : the number of pairs that are in the same partitions in both $\mathbf{U}$ and $\mathbf{V}$;
- (ii) N_{00} : the number of pairs that are in the different partitions in both $\mathbf{U}$ and $\mathbf{V}$;
- (iii) N_{01} : the number of pairs that are in the same partition in $\mathbf{U}$ but in different partitions in $\mathbf{V}$;
- (iv) N_{10} : the number of pairs that are in the different partitions in $\mathbf{U}$ but in the same partition in $\mathbf{V}$.

Thus, we can utilize the above notation to define ARI and NMI metrics (Yu et al., 2022; Vinh et al., 2009)

$$ARI(\mathbf{U}, \mathbf{V}) = \frac{2(N_{00}N_{11} - N_{01}N_{10})}{(N_{00} + N_{01})(N_{01} + N_{11}) + (N_{00} + N_{10})(N_{10} + N_{11})}$$

$$NMI(\mathbf{U}, \mathbf{V}) = \frac{I(\mathbf{U}, \mathbf{V})}{H(\mathbf{U}) + H(\mathbf{V})}$$

where $I(\mathbf{U}, \mathbf{V})$ is the mutual information between $\mathbf{U}$ and $\mathbf{V}$, denoted as

$$I(\mathbf{U}, \mathbf{V}) = \sum_{i=1}^R \sum_{j=1}^C \frac{|\mathcal{U}_i \cap \mathcal{V}_j|}{N} \log_2 \frac{N |\mathcal{U}_i \cap \mathcal{V}_j|}{|\mathcal{U}_i| |\mathcal{V}_j|}$$

and $H(\cdot)$ is the entropy of partitions, with $H(\mathbf{U})$ and $H(\mathbf{V})$ being calculated

$$H(\mathbf{U}) = - \sum_{i=1}^R \frac{|\mathcal{U}_i|}{N} \log \frac{|\mathcal{U}_i|}{N}$$

$$H(\mathbf{V}) = - \sum_{j=1}^C \frac{|\mathcal{V}_j|}{N} \log \frac{|\mathcal{V}_j|}{N}$$

At the end, for each simulation, we compute the ARI and NMI evaluation metrics using the above formula. By aggregating the results from all 20 simulations, we construct a boxplot for each clustering method to assess their performance across three simulated scenarios.

3.3.4 Results

In the subsequent analysis, we will examine the experimental results. A key finding has been summarized for each result.

Result 1: BayesSpace performs the best, K-means demonstrates the worst performance in the circle pattern

Figure 3.9 provides an evaluation of different model performances by comparing both ARI and NMI metrics. Overall, we can observe that BayesSpace performed the best, while K-means performed the worst. The clustering methods Hierarchical, Louvain, BANKSY, CARD, and RCTD showed similar performances. The excellent performance of BayesSpace indicates that spatial clustering by incorporating spatial coordinates can enhance the accuracy of detecting spatial domains. However, what surprised us was the bad performance of HMRF, though it was slightly better than K-means clustering. Subsequently, we conducted a detailed analysis to understand why the performance of these two algorithms was inferior to that of other algorithms. We found that the HMRF spatial clustering algorithm employs the BinSpect approach based on K-means to identify genes with a spatially coherent expression profile. As a result, this algorithm's performance is comparable to that of K-means.

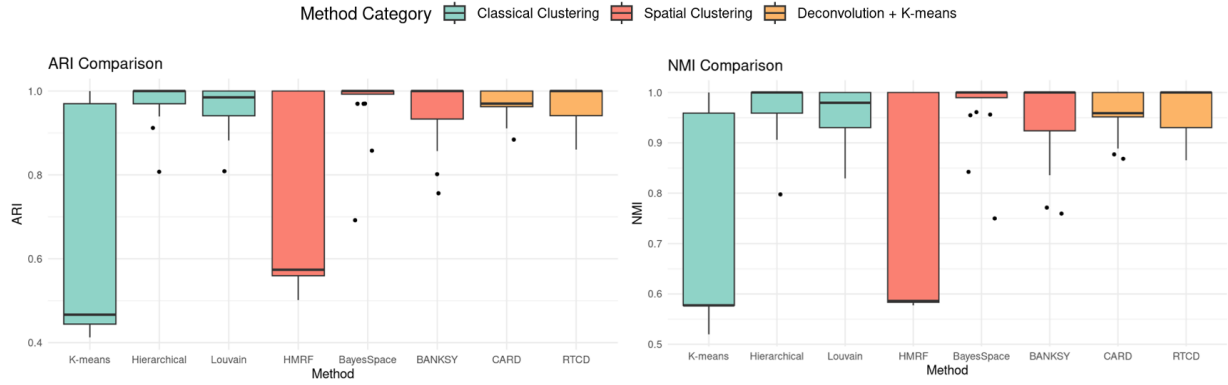


Figure 3.9: Summary of clustering accuracy in the circle pattern

In the boxplot, the center line, box limits, and whiskers denote the median, upper and lower quartiles, and 1.5x interquartile range, respectively

Result 2: By investigating the causes of high variability observed in the K-means algorithm, we discovered that good simulations present discernible regions, while bad simulations lack clear distinctions

Furthermore, we were intrigued by why K-means performance was the poorest. Theoretically, the performance of K-means should be similar to that of the Hierarchical clustering algorithm.

Drawing insights from the visualization of the simulated data's first two PCs, as depicted in Figure 3.10, we observed that in datasets where K-means achieved only an ARI of 0.4, the PCA-reduced representation did not show any clear patterns and seemed quite scattered. In particular, clusters 2 and 3 are mixed, lacking distinct separation, suggesting the clustering group is not apparent. On the contrary, in datasets where K-means had an ARI of 1, the Principal Component Analysis (PCA) efficiently discerned discernible regions, which enabled the K-means algorithm to distinguish between the three circle layers accurately.

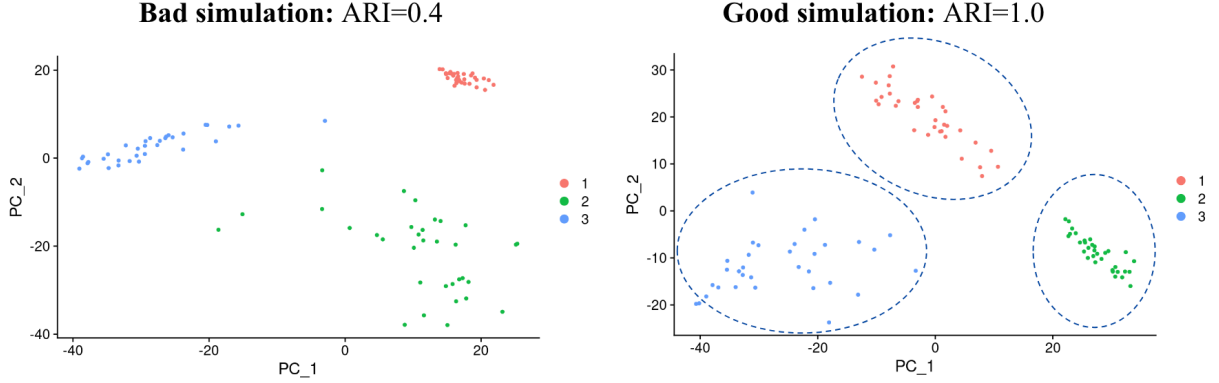


Figure 3.10: Comparison of two PCs in bad and good simulations

Result 3: K-means again shows the poorest performance, and no significant difference is observed among other clustering algorithms except for HMRF, which presents the best performance in region pattern

Performance evaluation results for the region pattern are depicted in Figure 3.11. From (a), we observe that K-means once again underperforms with a notably wide boxplot, indicating its performance is less consistent and generally inferior to the other methods. Therefore, to conduct a comparative analysis of the performance of other algorithms, we decided to exclude the K-means algorithm from the comparison, and the outcomes are presented in (b).

Surprisingly, the HMRF spatial clustering technique stands out in this region pattern, consistently recognizing the true spatial domain labels in every simulation, achieving perfect ARI and NMI scores of 1. However, in the first pattern, HMRF performs poorly, ranking second to last. As for the other algorithms, they exhibit very comparable performance, having a similar range and median. Furthermore, the deconvolution-based methods exhibit fewer outliers than the spatial clustering methods, implying greater stability of these methods in this pattern.

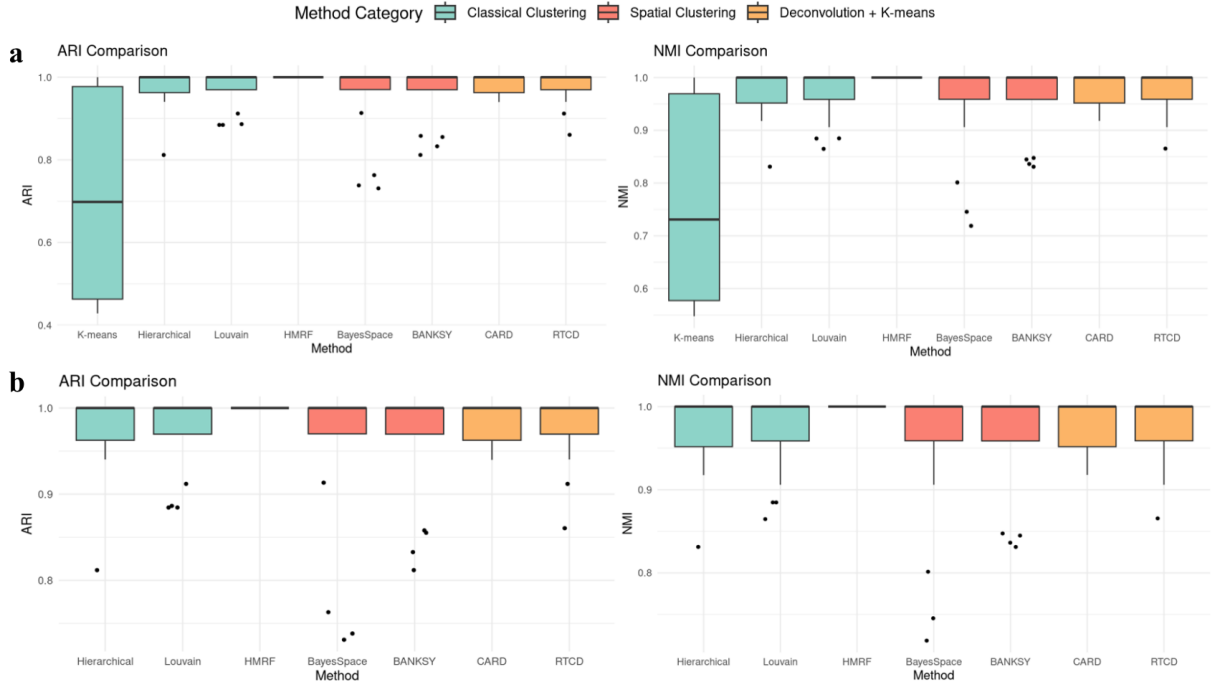


Figure 3.11: Summary of clustering accuracy in region pattern: (a) Comparison of all Methods; (b) Comparison excluding the K-means algorithm

In the boxplot, the center line, box limits, and whiskers denote the median, upper and lower quartiles, and 1.5x interquartile range, respectively

Result 4: By using patterns from real data in CARD simulation, we conclude that HMRf exhibits superior performance, while the effectiveness of other methods is approximately the same

The results are displayed in Figure 3.12. The HMRf method performs better than other algorithms, whereas the remaining algorithms exhibit similar performances, except for the K-means clustering method, which demonstrates the worst performance. It suggests that spatial clustering can enhance spatial domain detection accuracy.

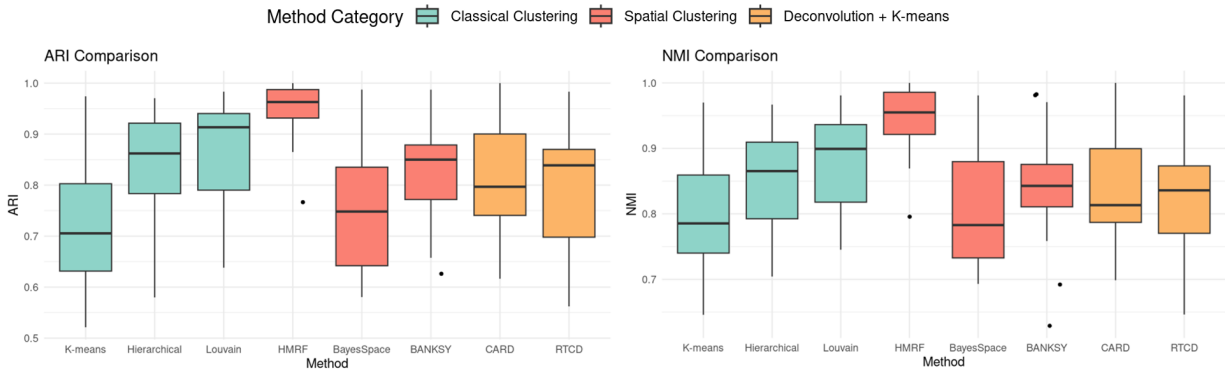


Figure 3.12: Summary of clustering accuracy in CARD simulation

In the boxplot, the center line, box limits, and whiskers denote the median, upper and lower quartiles, and 1.5x interquartile range, respectively

However, The median of boxplots for Louvain community detection and Hierarchical clustering is better than other spatial clustering algorithms. Furthermore, regarding the deconvolution-based method, RCTD is superior to CARD, but it is not apparent that incorporating cell type information improves the accuracy of spatial domain detection.

Result 5: By using patterns from real data in scDesign3 simulation, we found that HMRF outperforms while BANKSY also demonstrates excellent performance in the ARI metric

The outcomes are presented in Figure 3.13. We observe that apart from the spatial clustering BayesSpace algorithm, the spreads of other algorithms are very small, suggesting stable and consistent outcomes. Among spatial clustering algorithms, HMRF and BANKSY methods present better performance compared to classical clustering methods. Notably, the BANKSY algorithm performs best, achieving the highest ARI in one of the simulations. The deconvolution-based methods exhibit lower accuracy when compared to classical clustering, which was unexpected. However, when evaluating the NMI, we discover that deconvolution-based algorithms perform better than classical clustering.

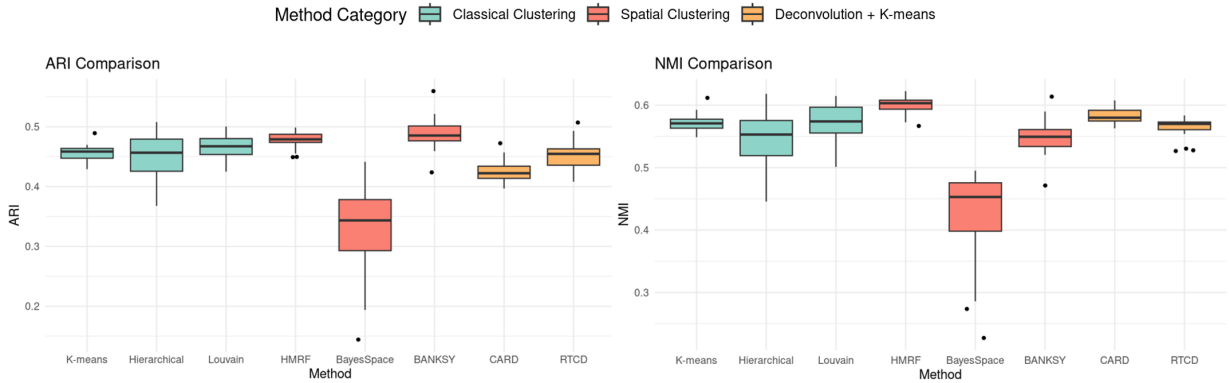


Figure 3.13: Summary of clustering accuracy in scDesign3 simulation
In the boxplot, the center line, box limits, and whiskers denote the median, upper and lower quartiles, and 1.5x interquartile range, respectively

Results 6: Simulations that account for spatial correlation demonstrate more challenges in spatial domain detection, exhibiting lower accuracy and narrower spread

In the subsequent analysis, we will do a comparative evaluation of the performance of CARD simulation and scDesign3 simulation to discern any differences between them, as depicted in Figure 3.14.

We observe that the overall ARI and NMI boxplots for each clustering method in the CARD simulation are higher compared to the scDesign3 simulation. Regarding the spread of boxplots, it can be noted that the scDesign3 simulation had a smaller spread than the

CARD simulation. We conjecture that the underlying reason can be attributed to the inherent characteristics of the simulation process. CARD simulation operates in a non-spatially correlated manner, whereas each spot in the scDesign3 simulation is spatially correlated. Hence, the scDesign3 simulation results are closer to real-world spatially resolved transcriptomics data due to incorporating a considerable amount of spatial information.

Nevertheless, including additional spatial information appears to lead to a decline in overall accuracy across all algorithms, making the precise identification of spatial domains more challenging. Furthermore, the smaller boxplot spread of the scDesign3 simulation indicates more stable and consistent outcomes, suggesting that the results are more reliable compared to CARD simulation.

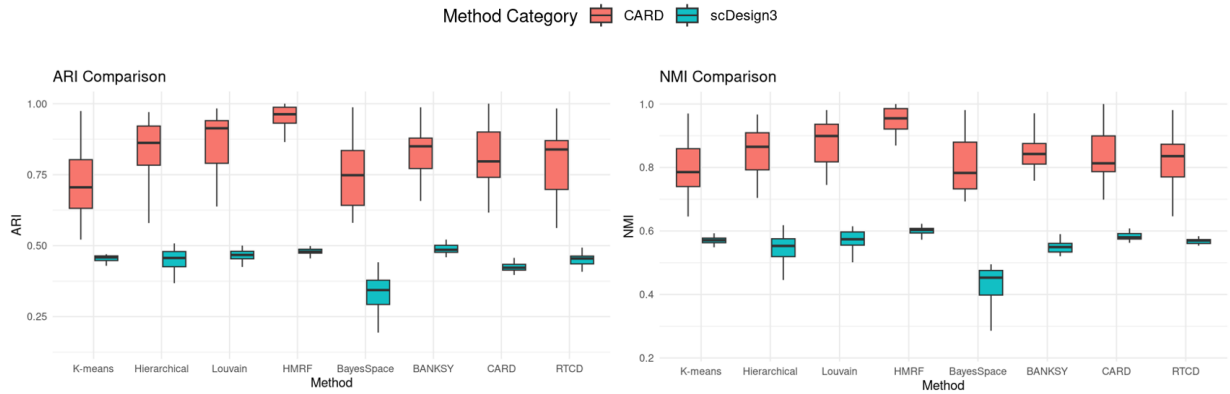


Figure 3.14: Evaluation comparison between two simulation methods

In the boxplot, the center line, box limits, and whiskers denote the median, upper and lower quartiles, and 1.5x interquartile range, respectively

3.4 Summary

In this chapter, we conducted a comparative analysis of two simulation patterns and explored a real dataset using two simulation methods. By comparing the performance of various clustering methods, we discovered that spatial clustering indeed improved the identification of spatial domains in specific simulation scenarios. However, in some scenarios, it may not perform as effectively as classical clustering. Additionally, we observed that utilizing realistic simulation methods provides us with more reliable and consistent results that are closer to reality.

Chapter 4

2D vs 3D Spatial Clustering Comparison

In this chapter, our primary focus is on investigating the impact of 3D information. We aim to explore whether the incorporation of 3D coordinates in clustering techniques enhances spatial domain detection performance compared to methods relying solely on 2D coordinates. To study the effects of 3D information, we will use the human dorsolateral prefrontal cortex (DLPFC) dataset, which offers the advantage of manually annotated ground truth for direct real-data analysis.

Within this chapter, we introduce two new 3D algorithms: the STitch3D and BASS methods. These advanced methods can simultaneously model data across multiple slices, enabling 3D reconstruction.

The chapter is structured to provide a detailed description of the real data in Section 4.1, followed by an in-depth explanation of the 3D reconstruction process using the STitch3D model in Section 4.2. We then outline our experimental design in Section 4.3. Subsequently, we conduct the experiments and present the key results in Section 4.4. Finally, the chapter concludes by summarizing the key findings and insights in Section 4.5.

4.1 Data description

DLPFC dataset Using the 10x Genomics Visium platform, spatial maps of gene expression in the adult human brain’s six-layered dorsolateral prefrontal cortex (DLPFC) were generated. Figure 4.1 shows the source of this experimental data. In Figure (a), the acquisition of DLPFC tissue blocks is in the anatomical plane, positioned perpendicularly to the pial surface and extending to the grey–white matter junction, with each block including all six cortical layers and White Matter (WM). Figure (b) outlines the experimental design schema, which consists of two sets of spatial replicates obtained from three distinct neurotypical adult donors. Every set comprises two consecutive and 10- μm tissue sections that are pretty close to each other, where the second set is situated 300 μm posterior to the initial set, yielding 12 samples that have been processed using the Visium platform. In this project, we mainly focus on analyzing the third subject, which is sample 151673-151676 ([Maynard et al., 2021](#)).

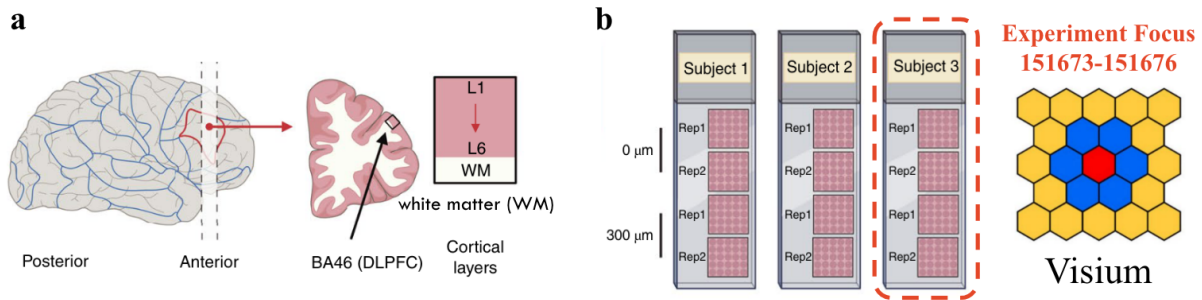


Figure 4.1: Spatial transcriptomics in DLPFC using Visium

Source: Maynard, K. R., Collado-Torres, L., Weber, L. M., Uyttingco, C., Barry, B. K., Williams, S. R., ... & Jaffe, A. E. (2021). Transcriptome-scale spatial gene expression in the human dorsolateral prefrontal cortex. *Nature neuroscience*, 24(3), 425-436. <https://doi.org/10.1038/s41593-020-00787-0>

Ground truth For the DLPFC dataset, the ground truth is acquired through manual annotation. Figure 4.2 illustrates how to obtain the ground truth. The left figure presents the tissue block of DLPFC and the histology derived from sample 151673. Then, seven distinct regions can be manually annotated. From the figure on the right-hand side, it can be seen that the annotations are layer by layer with a clear hierarchical structure, signifying different spatial domains (Maynard et al., 2021).

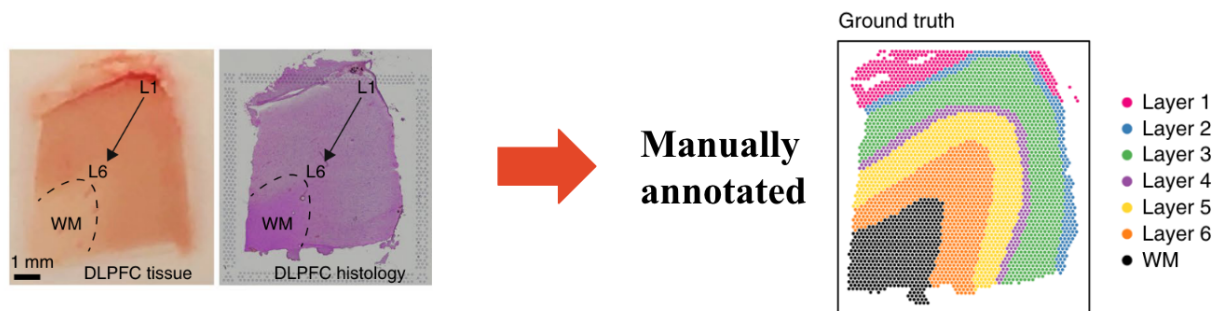


Figure 4.2: Ground truth for DLPFC dataset

Source: Maynard, K. R., Collado-Torres, L., Weber, L. M., Uyttingco, C., Barry, B. K., Williams, S. R., ... & Jaffe, A. E. (2021). Transcriptome-scale spatial gene expression in the human dorsolateral prefrontal cortex. *Nature neuroscience*, 24(3), 425-436. <https://doi.org/10.1038/s41593-020-00787-0>

4.2 3D reconstruction

In this section, we explored the methodology by which the STitch3D technique reconstructs the 3D structure. Figure 4.3 depicts this 3D reconstruction process. The preprocessing procedures of STitch3D encompassed the alignment of spots derived from multiple tissue slices to establish the 3D coordinates of these spots, as shown in Figure (a). In this dataset, four slices were selected and stacked to obtain the Z coordinate. After that, a comprehensive 3D graph was constructed as part of this process. Following this, the provided input was processed by the STitch3D model modelling, which employed a graph attention-based neural

network. The output generated by STitch3D provided information regarding the cell type proportion for each point. Then, by applying the Louvain clustering algorithm, we obtained results for 2D spatial domain detection, as demonstrated in Figure (b). The 3D reconstruction presented can be accomplished by stacking many slices together, utilising the results obtained from the 2D spatial domain detection. From Figure (c), we can observe that the 3D reconstruction results of STitch3D closely align with the manually annotated dataset, indicating that it can accurately identify most spatial domains in 3D tissue structures. Let us closely examine STitch3D's learned representations of spots generated by UMAP plots, and we can find that the clustering results and layer annotation are strikingly similar, as represented in Figure (d). In summary, the results indicate that STitch3D exhibits exceptional proficiency in reconstructing 3D structures, which is vital for interpreting meaningful biological tissue structures.

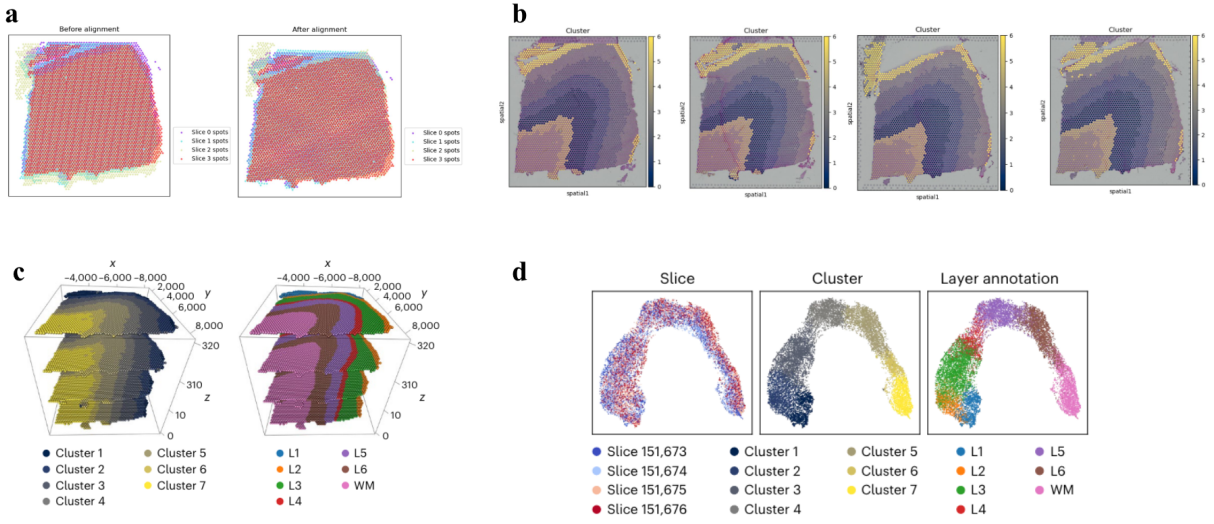


Figure 4.3: STitch3D: 3D reconstruction

a, Alignment multiple slices. **b**, STitch3D's 2D spatial domain detection result. **c**, STitch3D's 3D spatial domain detection result. **d**, UMAP plots present STitch3D's learned representations of spots within a shared latent space.

4.3 Experimental design

Figure 4.4 displays the experimental design of this chapter. The input for the experiment is a manually annotated DLPFC dataset. The input for classical clustering and 2D spatial clustering is a single slice. On the other hand, for 3D spatial clustering, multiple slices are used as the input. Subsequently, a preprocessing step was conducted. After completing the necessary preprocessing processes, our research shifted its focus to the central aspect: spatial domain detection. The methodology was categorized into three major classifications:

1. **Classical clustering:** By only using spatial expression data, we implemented three methods: K-means, Hierarchical clustering, and Louvain community detection.

2. **2D spatial clustering:** Within this particular category, there exist two distinct subdivisions. One involves spatial expression with the addition of x and y coordinates, which includes BayesSpace, HMRF, and BANKSY. In addition to incorporating spatial expression and x and y coordinates, the deconvolution technique also includes cell type information for spatial clustering, such as CARD and RCTD.
3. **3D spatial clustering:** This category can carry out joint modelling of multiple slices, perform 3D reconstruction, and then proceed with the spatial domain detection task. In this category, there are two advanced methods, named the BASS and STitch3D techniques.

Lastly, we can evaluate the performance of each method using the ARI and NMI metrics by comparing the spatial domain labels obtained from each clustering method to the manually annotated ground truth.

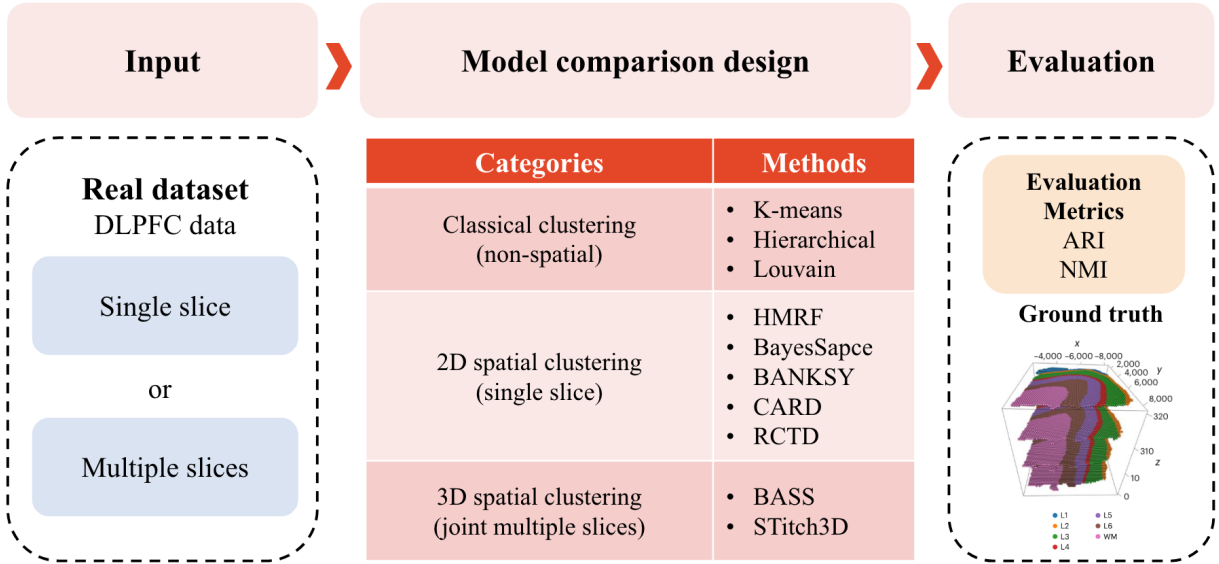


Figure 4.4: Overview of Experimental Design

4.4 Results

In this section, we will closely review the results from various clustering models. At first, we selected a few representative clustering methods from spatial domain detection results for a deeper investigation. After that, we computed evaluation metrics to assess the performance of three distinct clustering categories.

Result 1: Visually, 3D spatial domain detection appears to exhibit better performance compared to 2D spatial domain detection

In the following part, let us focus on the results of the spatial domain detection. We have chosen to compare the K-means from classical clustering, BayesSpace from 2D spatial clustering, and STitch3D and BASS from 3D spatial clustering. The outcomes of spatial domain detection are presented in Figure 4.5.

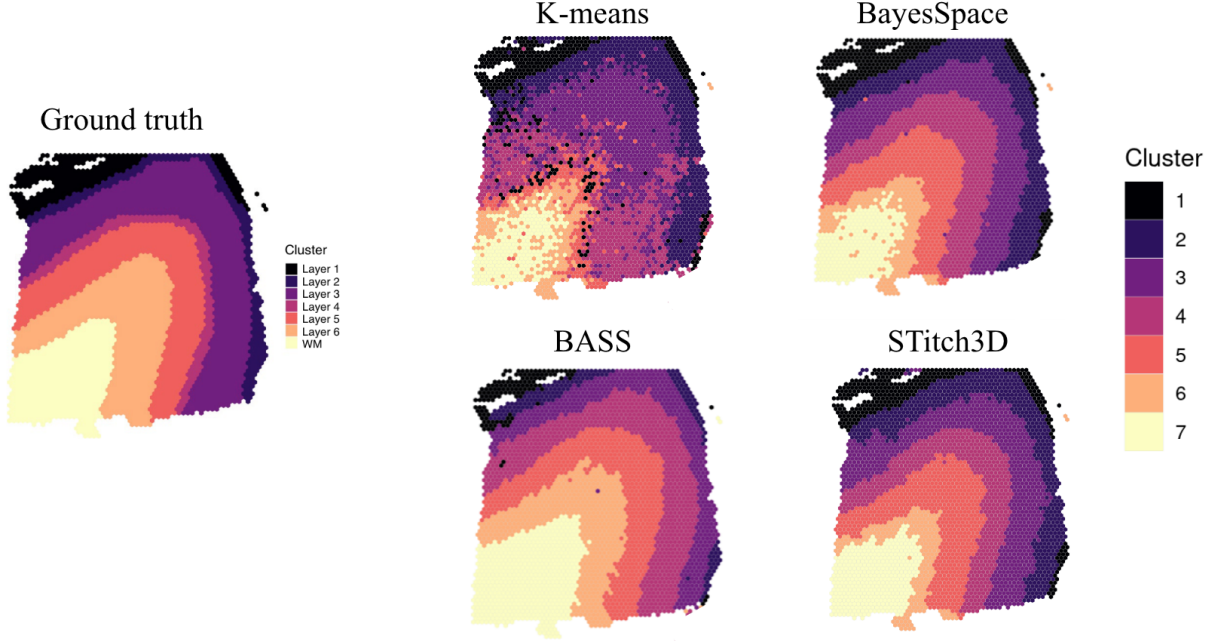


Figure 4.5: Spatial domain detection results

The above figure compares the results of the four spatial domain detection methods with the ground truth. The K-means clustering algorithm needs to identify the seven distinct regions accurately since many spots are misclassified. After integrating spatial coordinates information, BayesSpace performs notably better than K-means, identifying most spatial regions. Furthermore, when advanced 3D reconstruction information is integrated, both BASS and STitch3D enhance the accuracy of identifying spatial regions compared to only utilizing 2D spatial information, closely coinciding with the ground truth.

Result 2: Quantitatively, 3D spatial domain detection appears to exhibit better performance compared to 2D spatial domain detection

In the subsequent analysis, the performance of various clustering methods is assessed by employing the ARI and NMI metrics. In particular, we focused on sample 151673. The evaluation result is depicted in Figure 4.6.

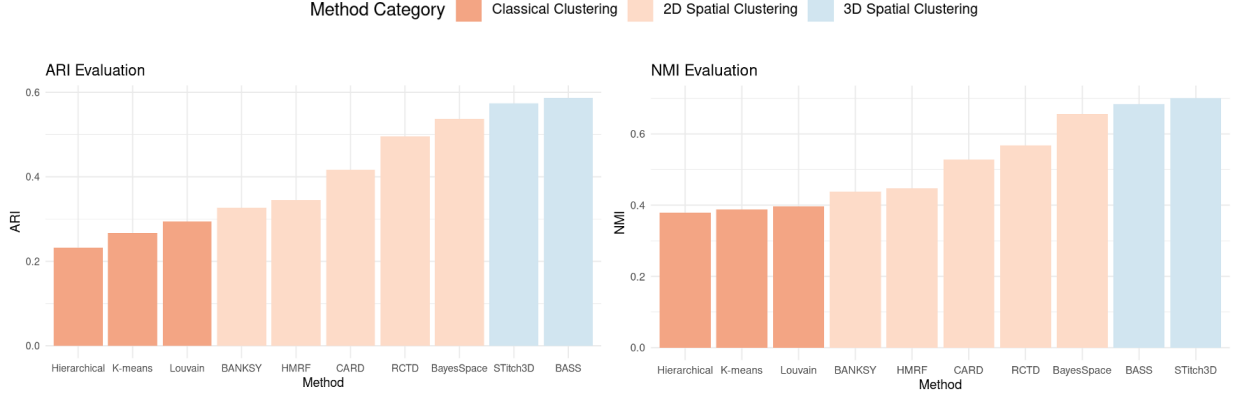


Figure 4.6: Summary of clustering accuracy in DLPFC dataset

Initially, we conduct a preliminary comparison between the results of ARI and NMI. We can observe that the rankings across three categories of clustering methods for both metrics display significant similarity. The classical clustering method, which does not incorporate spatial information, exhibits the poorest performance. In contrast, 3D spatial clustering methods utilizing 3D coordinates outperform the 2D spatial clustering methods only using 2D coordinates. When closely investigating the rankings of specific methods, most are consistent, except for BASS and STitch3D approaches. In ARI evaluation, the BASS technique is better than STitch3D. However, the STitch3D method is superior to BASS when evaluating using the NMI metric.

In conclusion, the 3D spatial clustering techniques performed better than the 2D spatial clustering techniques, suggesting the importance of 3D reconstructions by jointly modelling multiple slices, as they effectively improve spatial domain recognition performance and enable more precise identification of spatial regions in the tissue section.

4.5 Summary

In this chapter, we have demonstrated that the 3D spatial clustering technique can enhance accuracy when performing spatial domain detection tasks. Accurately conducting 3D tissue reconstruction is an effective means of capturing and representing the significant structures of a tissue or organism. The advanced techniques employed for detecting spatial domains in three dimensions hold significant meaning.

Chapter 5

Conclusion

5.1 Discussion

Identifying spatial domains is a crucial question in analysing spatially resolved transcriptomics data, which is a complex yet important challenge (Teng et al., 2022). In this thesis, we compare various clustering methods, ranging from traditional scRNA-seq clustering approaches to the latest spatial clustering methods that incorporate spatial 2D or 3D coordinates. Through comprehensive comparisons using both simulated and real data, we derived certain conclusions. In specific situations, employing spatial clustering indeed enhances the precision of spatial domain detection.

We acknowledge several limitations in our study. Firstly, we employed two simulation techniques in Chapter 3. There are only 20 simulation times for each scenario, suggesting small sample sizes. Given the limited number of simulations, the results depicted in the evaluation boxplots might be influenced by bad simulations, referred to as outliers. Such outliers might skew the overall clustering performance, potentially leading to inaccurate conclusions.

In addition, another limitation we observed is unstable performance. Certain clustering algorithms might exhibit instability. There are instances where spatial clustering performs better than classical clustering, but there exist some cases where it underperforms. The inconsistency in the results precludes us from drawing a consistent conclusion.

Furthermore, within the scope of our experiments, our analysis only utilised two real datasets, implying a notable limitation. If we do not extend the scope of our validation to a broader range of real-world datasets, we may miss capturing diverse real datasets or challenges in different spatially resolved transcriptomics data. Consequently, this restricted dataset selection might introduce bias into our conclusions and limit the generalizability of our findings to the broader realm of spatially resolved transcriptomics.

In light of the limitations mentioned above, we propose a few directions for future research.

Data simulation - Increase the sample size, for instance, 100, 200, and so forth, to assess the impact of sample size on clustering performance. Moreover, consider more complex or biologically meaningful simulation scenarios for a more comprehensive study.

Stability investigation - Investigate the underlying reasons for some spatial clustering algorithmic instability. Suppose specific clustering methods indeed perform poorly in particular situations. In that case, improvements can be approached from the model design

perspective based on this context, aiming to design a model and optimising the model's performance to better adapt to diverse datasets.

Expand datasets - Consider exploring different types of spatially resolved transcriptomics datasets, applying our experimental design across different datasets for a more thorough validation of the performance of 2D or 3D spatial domain detection methods.

Apart from using evaluation metrics to assess the performance of different clustering algorithms, we can also consider sensitivity, scalability, running speed, and running time as indicators for evaluating the performance of the clustering algorithms.

The design of our experiment can also be expanded. After undertaking experiments with various simulations or real datasets, we can attempt to develop a supervised linear model to explore what influencing factors affect the performance of clustering specifically.

5.2 Conclusion

In summary, this thesis provides an in-depth review and comparison of spatial clustering methods in spatially resolved transcriptomics data. Through simulation and real data analysis, it is evident that different spatial clustering methods perform differently on various datasets. Some algorithms exhibit better performance in incorporating spatial information. Additionally, the comparison of spatial clustering methods gains greater significance when a more realistic simulation approach is employed, as the choice of simulation methods can significantly impact our conclusions regarding clustering performance. By utilizing more realistic simulation methods, such as scDesign3, it is more likely to infer that spatial clustering outperforms traditional clustering in terms of performance. Notably, 3D spatial clustering significantly enhances the accuracy of spatial domain detection compared to the implementation of 2D spatial clustering layer by layer, as the 3D technique can reconstruct biological structures to closely resemble real biological organisms or tissues.

We hope that the design and methodology of this study will significantly advance research in spatial domain detection. Through meticulous comparative analysis, our objective is to uncover nuanced insights and foster a deeper understanding of spatial clustering methodologies. This endeavor aims not only to elucidate the inherent strengths and weaknesses of various spatial clustering approaches but also to lay a robust foundation for subsequent investigations in this field. By thoroughly dissecting the intricacies of spatial clustering, we aspire to make a substantial contribution to the ongoing scholarly discourse and ultimately enhance the practical applicability of spatial clustering techniques in addressing real-world challenges.

Chapter 6

Appendix

6.1 Data availability

All data used in this work are publicly available through online sources:

1. **Human heart dataset** in Chapter 3:

- human embryonic heart dataset profiled by ST platform
(<https://data.mendeley.com/datasets/dgnysc3zn5/1>)
- human embryonic heart dataset profiled by 10x Genomics Chromium platform
(<https://data.mendeley.com/datasets/mbvhhf8m62/2>)

2. **human dorsolateral prefrontal cortex (DLPFC) dataset** used in Chapter 4:

- human dorsolateral prefrontal cortex dataset profiled by Visium platform
(<http://spatial.libd.org/spatialLIBD/>)
- human dorsolateral prefrontal cortex dataset profiled by 10x Genomics Chromium platform (GSE144136)
(<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE144136>)

6.2 Code availability

The codes in this project are available from the website: <https://github.com/cristalliao/Honours-Project>

6.3 Supplementary information

CARD simulation for real human heart dataset

The key steps of the CARD simulation are illustrated in Figure 6.1.

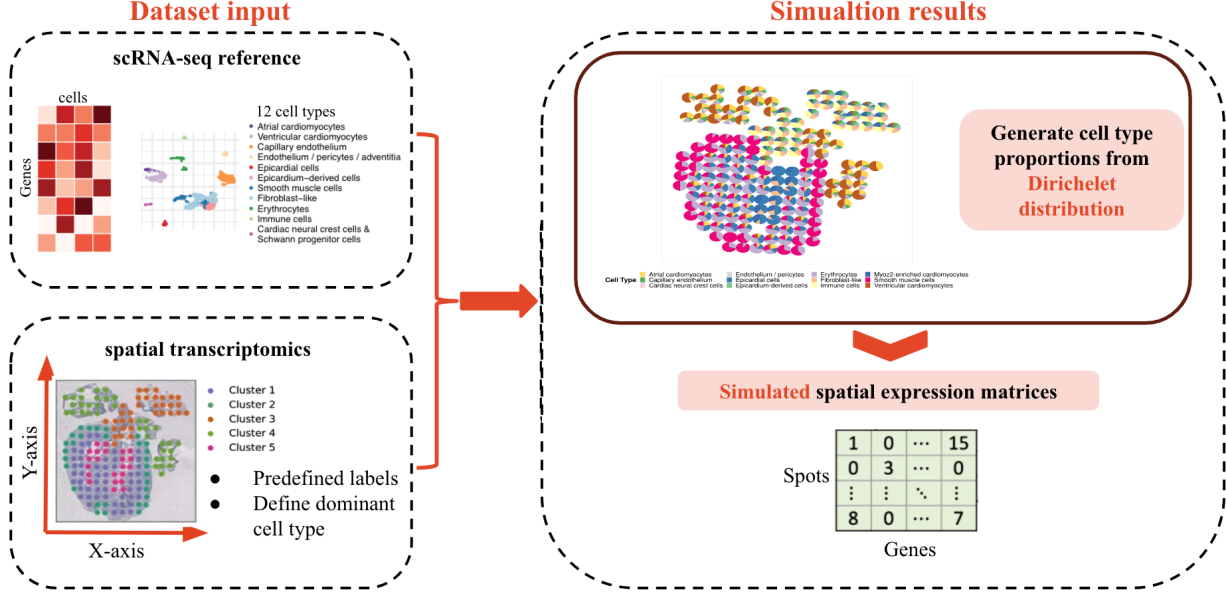


Figure 6.1: CARD simulation workflow

The two primary inputs are a scRNA-seq reference and a spatial transcriptomics dataset for the human heart dataset:

(1) **scRNA-seq reference:** This consists of a matrix that represents cells by genes and 12 different cell type information

(2) **spatial transcriptomics:** This dataset contains predefined labels for five spatial regions. Each spatial region has a designated dominant cell type. For instance, cluster 1 is tagged with the label “1”, where “Erythrocytes” was chosen as the dominant cell type for that cluster.

To initiate the simulation, we commenced by incorporating inputs into the procedure. Following this, we utilized the Dirichlet distribution to generate random proportions of cell types for each spot. Subsequently, these proportions were utilized as the foundation for generating a spot-by-gene spatial expression matrix. Through the CARD simulation, we can obtain cell type information in different spots, including both count and proportion. Also, the gene expression data for each spot was generated. However, it was important to note that the lack of spatial correlation in each spot due to the independence of the simulation process may lead to potential inaccuracies in capturing the true characteristics of the real dataset.

Bibliography

- Asp, Giacomello, S., Larsson, L., Wu, C., Fürth, D., Qian, X., Wärdell, E., Custodio, J., Reimegård, J., Salmén, F., et al., 2019. A spatiotemporal organ-wide gene expression and cell atlas of the developing human heart. *Cell* 179, 1647–1660.
- Bacher, R., Chu, L.F., Leng, N., Gasch, A.P., Thomson, J.A., Stewart, R.M., Newton, M., Kendzierski, C., 2017. Scnorm: robust normalization of single-cell rna-seq data. *Nature methods* 14, 584–586.
- Bacher, R., Kendzierski, C., 2016. Design and computational analysis of single-cell rna-sequencing experiments. *Genome biology* 17, 1–14.
- Bain, B.J., 2017. Structure and function of red and white blood cells. *Medicine* 45, 187–193.
- Banerjee, S., Carlin, B.P., Gelfand, A.E., 2014. Hierarchical modeling and analysis for spatial data. CRC press.
- Besag, J., 1986. On the statistical analysis of dirty pictures. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 48, 259–279.
- Brennecke, P., Anders, S., Kim, J.K., Kołodziejczyk, A.A., Zhang, X., Proserpio, V., Baying, B., Benes, V., Teichmann, S.A., Marioni, J.C., et al., 2013. Accounting for technical noise in single-cell rna-seq experiments. *Nature methods* 10, 1093–1095.
- Burgess, D.J., 2019. Spatial transcriptomics coming of age. *Nature Reviews Genetics* 20, 317–317.
- Cable, D.M., Murray, E., Zou, L.S., Goeva, A., Macosko, E.Z., Chen, F., Irizarry, R.A., 2022. Robust decomposition of cell type mixtures in spatial transcriptomics. *Nature biotechnology* 40, 517–526.
- Cao, J., O’Day, D.R., Pliner, H.A., Kingsley, P.D., Deng, M., Daza, R.M., Zager, M.A., Aldinger, K.A., Blecher-Gonen, R., Zhang, F., et al., 2020. A human cell atlas of fetal gene expression. *Science* 370, eaba7721.
- Cao, J., Spielmann, M., Qiu, X., Huang, X., Ibrahim, D.M., Hill, A.J., Zhang, F., Mundlos, S., Christiansen, L., Steemers, F.J., et al., 2019. The single-cell transcriptional landscape of mammalian organogenesis. *Nature* 566, 496–502.
- Chen, C., Kim, J.H., Yang, P., 2022. Evaluating spatially variable gene detection methods for spatial transcriptomics data. *bioRxiv* , 2022–11.

- Choe, K., Pak, U., Pang, Y., Hao, W., Yang, X., 2023. Advances and challenges in spatial transcriptomics for developmental biology. *Biomolecules* 13, 156.
- Cords, L., Tietscher, S., Anzeneder, T., Langwieder, C., Rees, M., de Souza, N., Bodenmiller, B., 2023. Cancer-associated fibroblast classification in single-cell and spatial proteomics data. *Nature Communications* 14, 4294.
- De Meo, P., Ferrara, E., Fiumara, G., Provetti, A., 2011. Generalized louvain method for community detection in large networks, in: 2011 11th international conference on intelligent systems design and applications, IEEE. pp. 88–93.
- DeBruine, Z.J., Melcher, K., Triche Jr, T.J., 2021. Fast and robust non-negative matrix factorization for single-cell experiments. *bioRxiv* , 2021–09.
- Dong, R., Yuan, G.C., 2021. Spatialdwl: accurate deconvolution of spatial transcriptomic data. *Genome biology* 22, 145.
- Dries, R., Zhu, Q., Dong, R., Eng, C.H.L., Li, H., Liu, K., Fu, Y., Zhao, T., Sarkar, A., Bao, F., et al., 2021. Giotto: a toolbox for integrative analysis and visualization of spatial expression data. *Genome biology* 22, 1–31.
- Fraley, C., Raftery, A.E., Murphy, T.B., Scrucca, L., 2012. mclust version 4 for R: normal mixture modeling for model-based clustering, classification, and density estimation. Technical Report. Technical report.
- Gottardo, R., Besag, J., Stephens, M., Murua, A., 2006. Probabilistic segmentation and intensity estimation for microarray images. *Biostatistics* 7, 85–99.
- Grün, D., Lyubimova, A., Kester, L., Wiebrands, K., Basak, O., Sasaki, N., Clevers, H., Van Oudenaarden, A., 2015. Single-cell messenger rna sequencing reveals rare intestinal cell types. *Nature* 525, 251–255.
- Haque, A., Engel, J., Teichmann, S.A., Lönnberg, T., 2017. A practical guide to single-cell rna-sequencing for biomedical research and clinical applications. *Genome medicine* 9, 1–12.
- Hebenstreit, D., 2012. Methods, challenges and potentials of single cell rna-seq. *Biology* 1, 658–667.
- Hu, J., Li, X., Coleman, K., Schroeder, A., Ma, N., Irwin, D.J., Lee, E.B., Shinohara, R.T., Li, M., 2021. Spagcn: Integrating gene expression, spatial location and histology to identify spatial domains and spatially variable genes by graph convolutional network. *Nature methods* 18, 1342–1351.

- Jones, A., Townes, F.W., Li, D., Engelhardt, B.E., 2022. Alignment of spatial genomics and histology data using deep gaussian processes. *BioRxiv* , 2022–01.
- Karlsson, M., Zhang, C., Méar, L., Zhong, W., Digre, A., Katona, B., Sjöstedt, E., Butler, L., Odeberg, J., Dusart, P., et al., 2021. A single-cell type transcriptomics map of human tissues. *Science advances* 7, eabh2169.
- Kiselev, V.Y., Andrews, T.S., Hemberg, M., 2019. Challenges in unsupervised clustering of single-cell rna-seq data. *Nature Reviews Genetics* 20, 273–282.
- Kiselev, V.Y., Kirschner, K., Schaub, M.T., Andrews, T., Yiu, A., Chandra, T., Natarajan, K.N., Reik, W., Barahona, M., Green, A.R., et al., 2017. Sc3: consensus clustering of single-cell rna-seq data. *Nature methods* 14, 483–486.
- Kleshchevnikov, V., Shmatko, A., Dann, E., Aivazidis, A., King, H.W., Li, T., Elmentaite, R., Lomakin, A., Kedlian, V., Gayoso, A., et al., 2022. Cell2location maps fine-grained cell types in spatial transcriptomics. *Nature biotechnology* 40, 661–671.
- L Lun, A.T., Bach, K., Marioni, J.C., 2016. Pooling across cells to normalize single-cell rna sequencing data with many zero counts. *Genome biology* 17, 1–14.
- Lacar, B., Linker, S.B., Jaeger, B.N., Krishnaswami, S.R., Barron, J.J., Kelder, M.J., Parylak, S.L., Paquola, A.C., Venepally, P., Novotny, M., et al., 2016. Nuclear rna-seq of single neurons reveals molecular signatures of activation. *Nature communications* 7, 11022.
- Lee, D., 2011. A comparison of conditional autoregressive models used in bayesian disease mapping. *Spatial and spatio-temporal epidemiology* 2, 79–89.
- Levine, J.H., Simonds, E.F., Bendall, S.C., Davis, K.L., El-ad, D.A., Tadmor, M.D., Litvin, O., Fienberg, H.G., Jager, A., Zunder, E.R., et al., 2015. Data-driven phenotypic dissection of aml reveals progenitor-like cells that correlate with prognosis. *Cell* 162, 184–197.
- Li, H., Zhou, J., Li, Z., Chen, S., Liao, X., Zhang, B., Zhang, R., Wang, Y., Sun, S., Gao, X., 2023. A comprehensive benchmarking with practical guidelines for cellular deconvolution of spatial transcriptomics. *Nature Communications* 14, 1548.
- Li, Z., Zhou, X., 2022. Bass: multi-scale and multi-sample analysis enables accurate cell type clustering and spatial domain detection in spatial transcriptomic studies. *Genome biology* 23, 168.
- Lin, P., Troup, M., Ho, J.W., 2017. Cidr: Ultrafast and accurate clustering through imputation for single-cell rna-seq data. *Genome biology* 18, 1–11.

- Lin, Y., 2022. Statistical modelling and machine learning for single cell data harmonisation and analysis. Phd thesis. University of Sydney. Sydney, AU. Available at <https://hdl.handle.net/2123/28034>.
- Lin, Y., Yang, J.Y., 2022. 3d reconstruction of spatial expression. *Nature Methods* 19, 526–527.
- Liu, W., Liao, X., Yang, Y., Lin, H., Yeong, J., Zhou, X., Shi, X., Liu, J., 2022. Joint dimension reduction and clustering analysis of single-cell rna-seq and spatial transcriptomics data. *Nucleic acids research* 50, e72–e72.
- Lloyd, S., 1982. Least squares quantization in pcm. *IEEE transactions on information theory* 28, 129–137.
- Ma, Y., Zhou, X., 2022. Spatially informed cell-type deconvolution for spatial transcriptomics. *Nature biotechnology* 40, 1349–1359.
- Marinov, G.K., Williams, B.A., McCue, K., Schroth, G.P., Gertz, J., Myers, R.M., Wold, B.J., 2014. From single-cell to cell-pool transcriptomes: stochasticity in gene expression and rna splicing. *Genome research* 24, 496–510.
- Maynard, K.R., Collado-Torres, L., Weber, L.M., Uytingco, C., Barry, B.K., Williams, S.R., Catallini, J.L., Tran, M.N., Besich, Z., Tippani, M., et al., 2021. Transcriptome-scale spatial gene expression in the human dorsolateral prefrontal cortex. *Nature neuroscience* 24, 425–436.
- McCarthy, D.J., Campbell, K.R., Lun, A.T., Wills, Q.F., 2017. Scater: pre-processing, quality control, normalization and visualization of single-cell rna-seq data in r. *Bioinformatics* 33, 1179–1186.
- Moses, L., Pachter, L., 2022. Museum of spatial transcriptomics. *Nature Methods* 19, 534–546.
- Newman, A.M., Liu, C.L., Green, M.R., Gentles, A.J., Feng, W., Xu, Y., Hoang, C.D., Diehn, M., Alizadeh, A.A., 2015. Robust enumeration of cell subsets from tissue expression profiles. *Nature methods* 12, 453–457.
- Petegrosso, R., Li, Z., Kuang, R., 2020. Machine learning and statistical methods for clustering single-cell rna-sequencing data. *Briefings in bioinformatics* 21, 1209–1223.
- Pham, D., Tan, X., Xu, J., Grice, L.F., Lam, P.Y., Raghubar, A., Vukovic, J., Ruitenberg, M.J., Nguyen, Q., 2020. stlearn: integrating spatial location, tissue morphology and gene expression to find cell types, cell-cell interactions and spatial trajectories within undissociated tissues. *BioRxiv* , 2020–05.

- Pierson, E., Yau, C., 2015. Zifa: Dimensionality reduction for zero-inflated single-cell gene expression analysis. *Genome biology* 16, 1–10.
- Rodriques, S.G., Stickels, R.R., Goeva, A., Martin, C.A., Murray, E., Vanderburg, C.R., Welch, J., Chen, L.M., Chen, F., Macosko, E.Z., 2019. Slide-seq: A scalable technology for measuring genome-wide expression at high spatial resolution. *Science* 363, 1463–1467.
- Satija, R., Farrell, J.A., Gennert, D., Schier, A.F., Regev, A., 2015. Spatial reconstruction of single-cell gene expression data. *Nature biotechnology* 33, 495–502.
- Shah, S., Lubeck, E., Zhou, W., Cai, L., 2016. In situ transcription profiling of single cells reveals spatial organization of cells in the mouse hippocampus. *Neuron* 92, 342–357.
- Shang, L., Zhou, X., 2022. Spatially aware dimension reduction for spatial transcriptomics. *Nature Communications* 13, 7203.
- Singhal, V., Chou, N., Lee, J., Liu, J., Chock, W.K., Lin, L., Chang, Y.C., Teo, E., Lee, H.K., Chen, K.H., et al., 2022. Banksy: a spatial omics algorithm that unifies cell type clustering and tissue domain segmentation. *bioRxiv* , 2022–04.
- Song, D., Wang, Q., Yan, G., Liu, T., Sun, T., Li, J.J., 2023. scdesign3 generates realistic in silico data for multimodal single-cell and spatial omics. *Nature Biotechnology* , 1–6.
- Ståhl, P.L., Salmén, F., Vickovic, S., Lundmark, A., Navarro, J.F., Magnusson, J., Giacomello, S., Asp, M., Westholm, J.O., Huss, M., et al., 2016. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. *Science* 353, 78–82.
- Su, K., Yu, T., Wu, H., 2021. Accurate feature selection improves single-cell rna-seq cell clustering. *Briefings in Bioinformatics* 22, bbab034.
- Sun, S., Zhu, J., Ma, Y., Zhou, X., 2019. Accuracy, robustness and scalability of dimensionality reduction methods for single-cell rna-seq analysis. *Genome biology* 20, 1–21.
- Sun, S., Zhu, J., Zhou, X., 2020. Statistical analysis of spatial expression patterns for spatially resolved transcriptomic studies. *Nature methods* 17, 193–200.
- Svensson, V., Teichmann, S.A., Stegle, O., 2018. Spatialde: identification of spatially variable genes. *Nature methods* 15, 343–346.
- Svensson, V., da Veiga Beltrame, E., Pachter, L., 2020. A curated database reveals trends in single-cell transcriptomics. *Database* 2020, baaa073.
- Tang, F., Barbacioru, C., Wang, Y., Nordman, E., Lee, C., Xu, N., Wang, X., Bodeau, J., Tuch, B.B., Siddiqui, A., et al., 2009. mrna-seq whole-transcriptome analysis of a single cell. *Nature methods* 6, 377–382.

- Teng, H., Yuan, Y., Bar-Joseph, Z., 2022. Clustering spatial transcriptomics data. *Bioinformatics* 38, 997–1004.
- Traag, V.A., Waltman, L., Van Eck, N.J., 2019. From louvain to leiden: guaranteeing well-connected communities. *Scientific reports* 9, 5233.
- Vinh, N.X., Epps, J., Bailey, J., 2009. Information theoretic measures for clusterings comparison: is a correction for chance necessary?, in: *Proceedings of the 26th annual international conference on machine learning*, pp. 1073–1080.
- Wagner, A., Regev, A., Yosef, N., 2016. Revealing the vectors of cellular identity with single-cell genomics. *Nature biotechnology* 34, 1145–1160.
- Wang, B., Zhu, J., Pierson, E., Ramazzotti, D., Batzoglou, S., 2017. Visualization and analysis of single-cell rna-seq data by kernel-based similarity learning. *Nature methods* 14, 414–416.
- Wang, G., Zhao, J., Yan, Y., Wang, Y., Wu, A.R., Yang, C., 2023a. Construction of a 3d whole organism spatial atlas by joint modeling of multiple slices. *bioRxiv* , 2023–02.
- Wang, M., Hu, Q., Lv, T., Wang, Y., Lan, Q., Xiang, R., Tu, Z., Wei, Y., Han, K., Shi, C., et al., 2022. High-resolution 3d spatiotemporal transcriptomic maps of developing drosophila embryos and larvae. *Developmental Cell* 57, 1271–1283.
- Wang, R., Peng, G., Tam, P.P., Jing, N., 2023b. Integration of computational analysis and spatial transcriptomics in single-cell studies. *Genomics, Proteomics & Bioinformatics* 21, 13–23.
- Waylen, L.N., Nim, H.T., Martelotto, L.G., Ramialison, M., 2020. From whole-mount to single-cell spatial assessment of gene expression in 3d. *Communications biology* 3, 602.
- Weber, L.M., Saha, A., Datta, A., Hansen, K.D., Hicks, S.C., 2023. nnsvg for the scalable identification of spatially variable genes using nearest-neighbor gaussian processes. *Nature communications* 14, 4059.
- Williams, C.G., Lee, H.J., Asatsuma, T., Vento-Tormo, R., Haque, A., 2022. An introduction to spatial transcriptomics for biomedical research. *Genome Medicine* 14, 1–18.
- Wolf, F.A., Angerer, P., Theis, F.J., 2018. Scanpy: large-scale single-cell gene expression data analysis. *Genome biology* 19, 1–5.
- Xiang, R., Wang, W., Yang, L., Wang, S., Xu, C., Chen, X., 2021. A comparison for dimensionality reduction methods of single-cell rna-seq data. *Frontiers in genetics* 12, 646936.

- Yang, Y., Shi, X., Liu, W., Zhou, Q., Chan Lau, M., Chun Tatt Lim, J., Sun, L., Ng, C.C.Y., Yeong, J., Liu, J., 2022. Sc-meb: spatial clustering with hidden markov random field using empirical bayes. *Briefings in bioinformatics* 23, bbab466.
- Yip, S.H., Sham, P.C., Wang, J., 2019. Evaluation of tools for highly variable gene discovery from single-cell rna-seq data. *Briefings in bioinformatics* 20, 1583–1589.
- Yip, S.H., Wang, P., Kocher, J.P.A., Sham, P.C., Wang, J., 2017. Linnorm: improved statistical analysis for single cell rna-seq expression data. *Nucleic acids research* 45, e179–e179.
- Yu, L., Cao, Y., Yang, J.Y., Yang, P., 2022. Benchmarking clustering algorithms on estimating the number of cell types from single-cell rna-sequencing data. *Genome biology* 23, 49.
- Yuan, G.C., Cai, L., Elowitz, M., Enver, T., Fan, G., Guo, G., Irizarry, R., Kharchenko, P., Kim, J., Orkin, S., et al., 2017. Challenges and emerging directions in single-cell analysis. *Genome biology* 18, 1–8.
- Zeira, R., Land, M., Strzalkowski, A., Raphael, B.J., 2022. Alignment and integration of spatial transcriptomics data. *Nature Methods* 19, 567–575.
- Zeisel, A., Muñoz-Manchado, A.B., Codeluppi, S., Lönnerberg, P., La Manno, G., Juréus, A., Marques, S., Munguba, H., He, L., Betsholtz, C., et al., 2015. Cell types in the mouse cortex and hippocampus revealed by single-cell rna-seq. *Science* 347, 1138–1142.
- Zhang, H., Lee, C.A., Li, Z., Garbe, J.R., Eide, C.R., Petegrosso, R., Kuang, R., Tolar, J., 2018. A multitask clustering approach for single-cell rna-seq analysis in recessive dystrophic epidermolysis bullosa. *PLoS computational biology* 14, e1006053.
- Zhao, E., Stone, M.R., Ren, X., Guenthoer, J., Smythe, K.S., Pulliam, T., Williams, S.R., Uytingco, C.R., Taylor, S.E., Nghiem, P., et al., 2021. Spatial transcriptomics at subspot resolution with bayesspace. *Nature biotechnology* 39, 1375–1384.
- Zhu, J., Sun, S., Zhou, X., 2021. Spark-x: non-parametric modeling enables scalable and robust detection of spatial expression patterns for large spatial transcriptomic studies. *Genome biology* 22, 1–25.
- Zhu, Q., Shah, S., Dries, R., Cai, L., Yuan, G.C., 2018. Identification of spatially associated subpopulations by combining scrnaseq and sequential fluorescence in situ hybridization data. *Nature biotechnology* 36, 1183–1190.
- Žurauskienė, J., Yau, C., 2016. pcreduce: hierarchical clustering of single cell transcriptional profiles. *BMC bioinformatics* 17, 1–11.